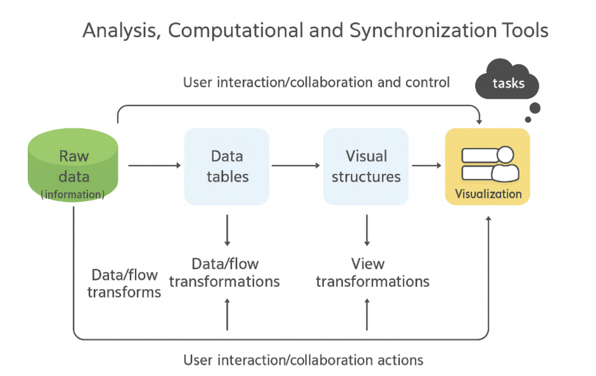
COURSE INFORMATION

DSK808 Visualization Models

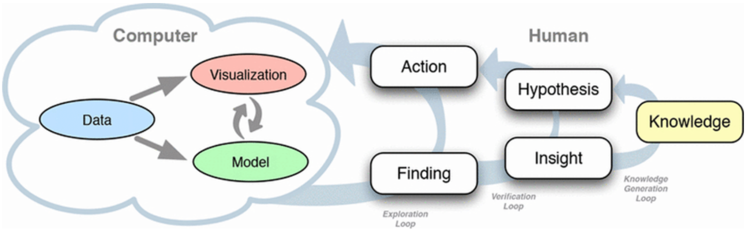
Fall 2025

Models for Visualization Design

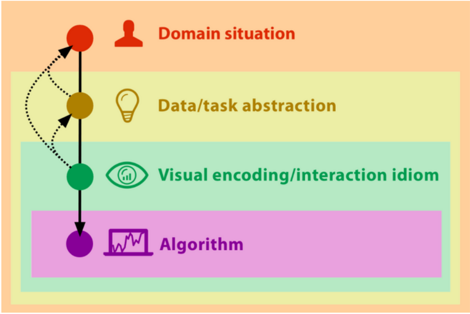
Visualization Pipeline



Knowledge Generation Model



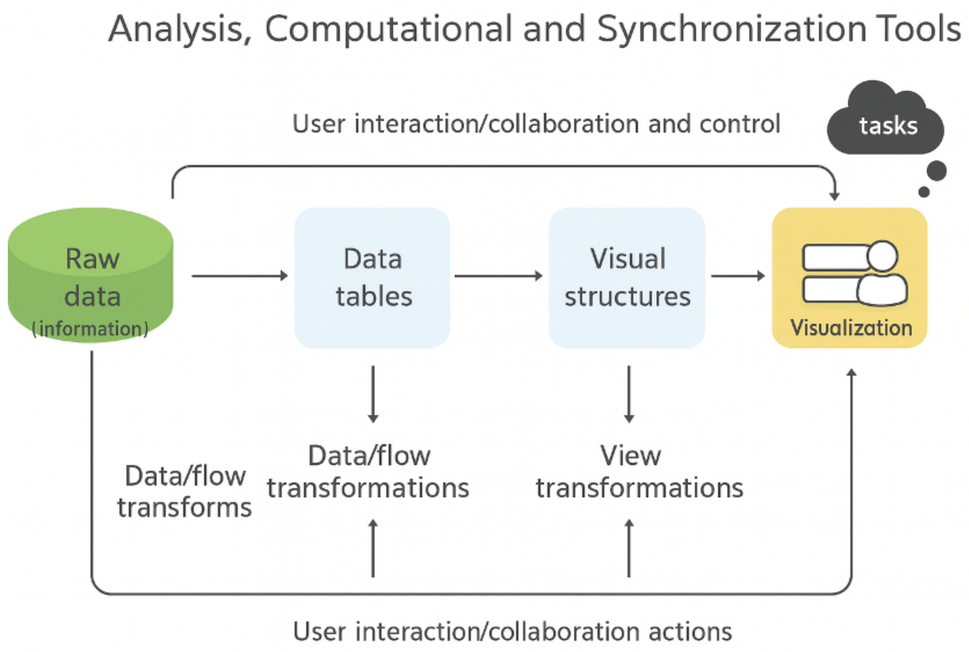
Nested Model



from Data

To Humans

Visualization Pipeline



Card, M. (1999). *Readings in information visualization: using vision to think*. Morgan Kaufmann.
Moreland, K. (2012). A survey of visualization pipelines. IEEE transactions on visualization and computer graphics, 19(3), 367-378.

Visualization Pipeline

- **Raw Data:** idiosyncratic formats
- **Data Tables:** meta-data and relations, processible for visualization purposes
- **Visual Structures:** generic visual representation mechanisms like charts with their visual properties
- **Views:** renderings of visual structures, different views show different parts of the visual structures

Argentina | Ranked #89

Life Expectancy: 78 Years

South America

Average: 76 Years
Minimum: 69 Years
Maximum: 82 Years

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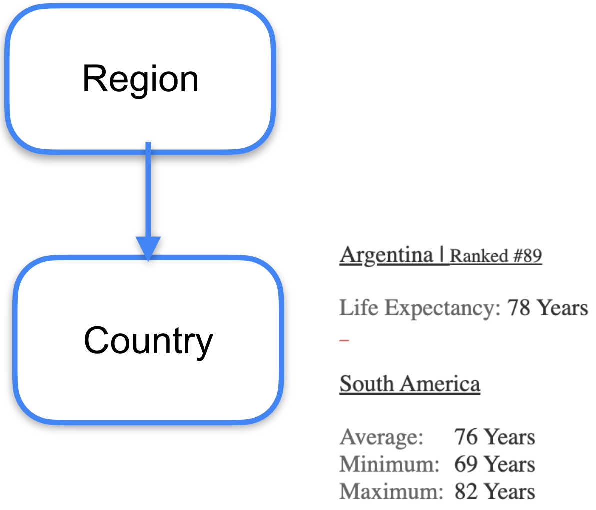

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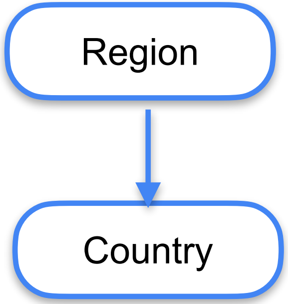

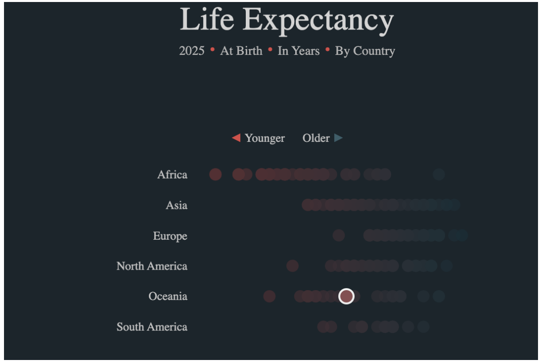

https://public.tableau.com/app/profile/luke.abraham/viz/LifeExpectancyMakeoverMonday/LifeExpectancy

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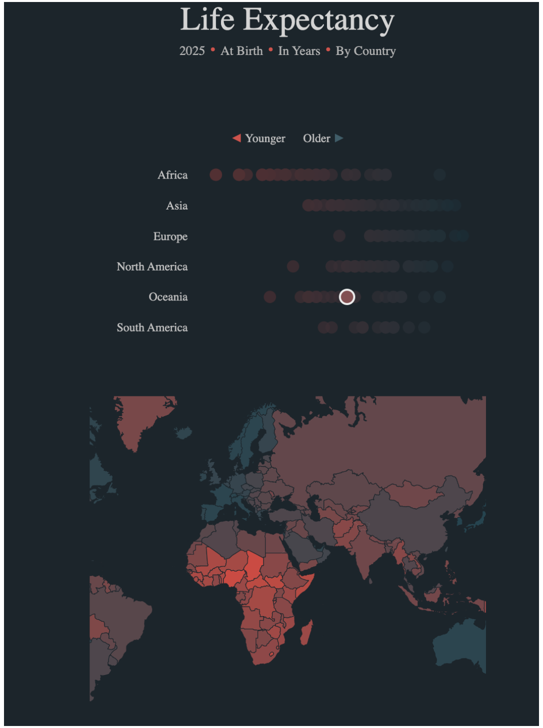

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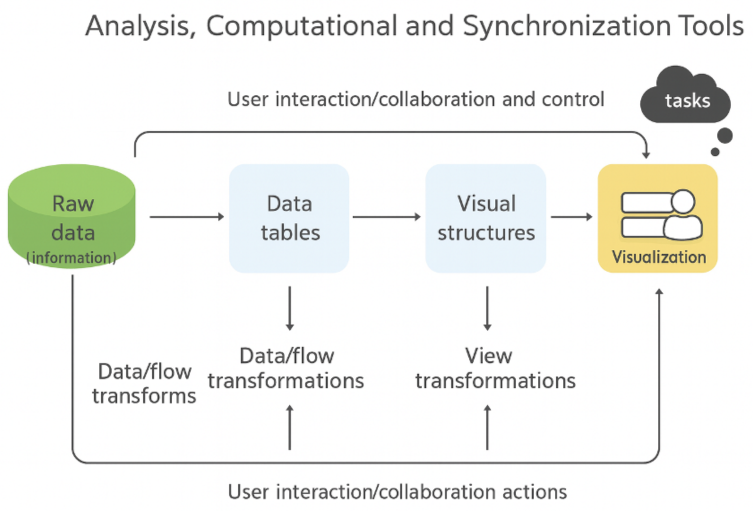
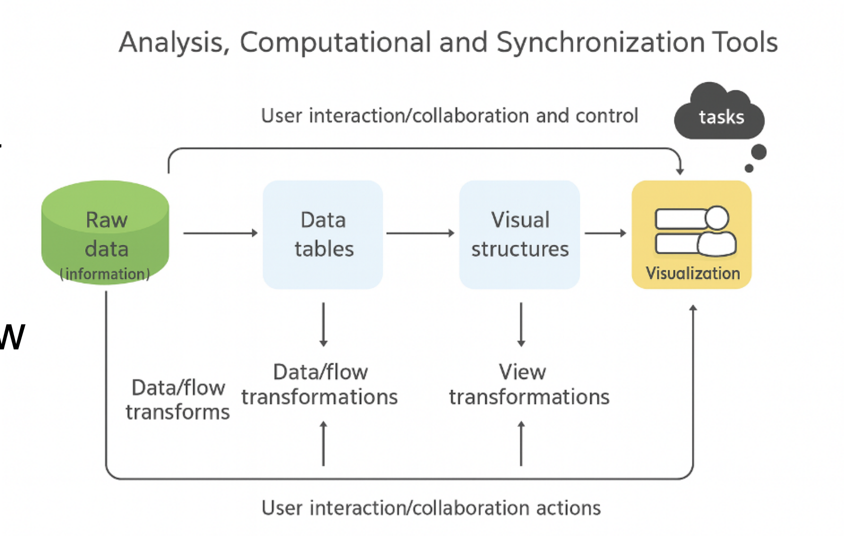


Visualization Pipeline

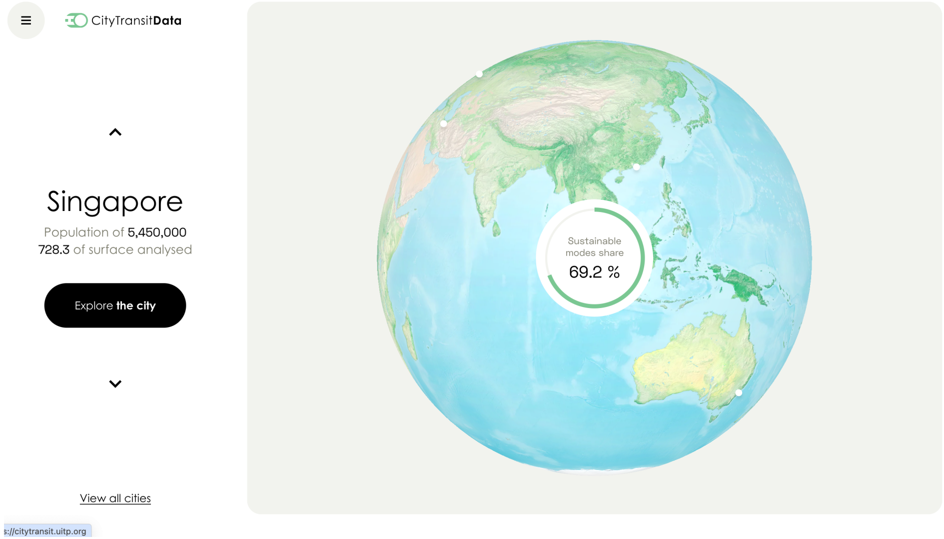
- **Data Transformations:** Mapping raw data into an organization fit for visualization
- **Visual Mappings:** Encoding abstract data into a visual representation
- **View Transformations:** Changing the view or perspective onto the visual representation



Information Seeking Mantra

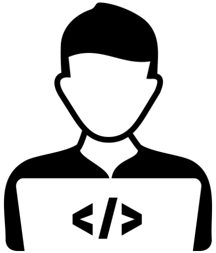
- "Overview first, zoom and filter, then details-on-demand"

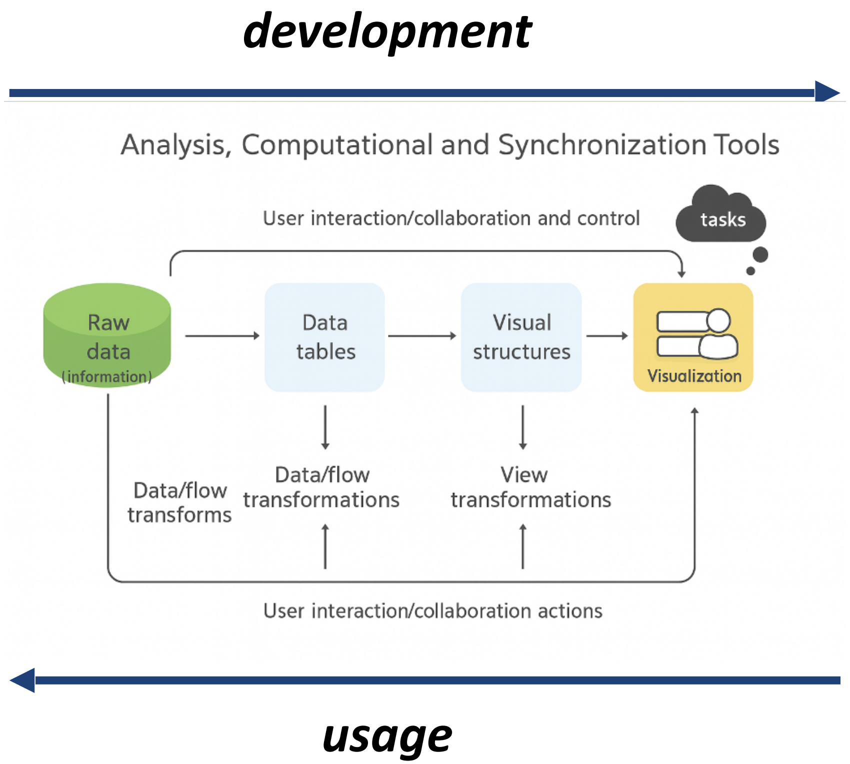
  by Ben Shneiderman in 1996.



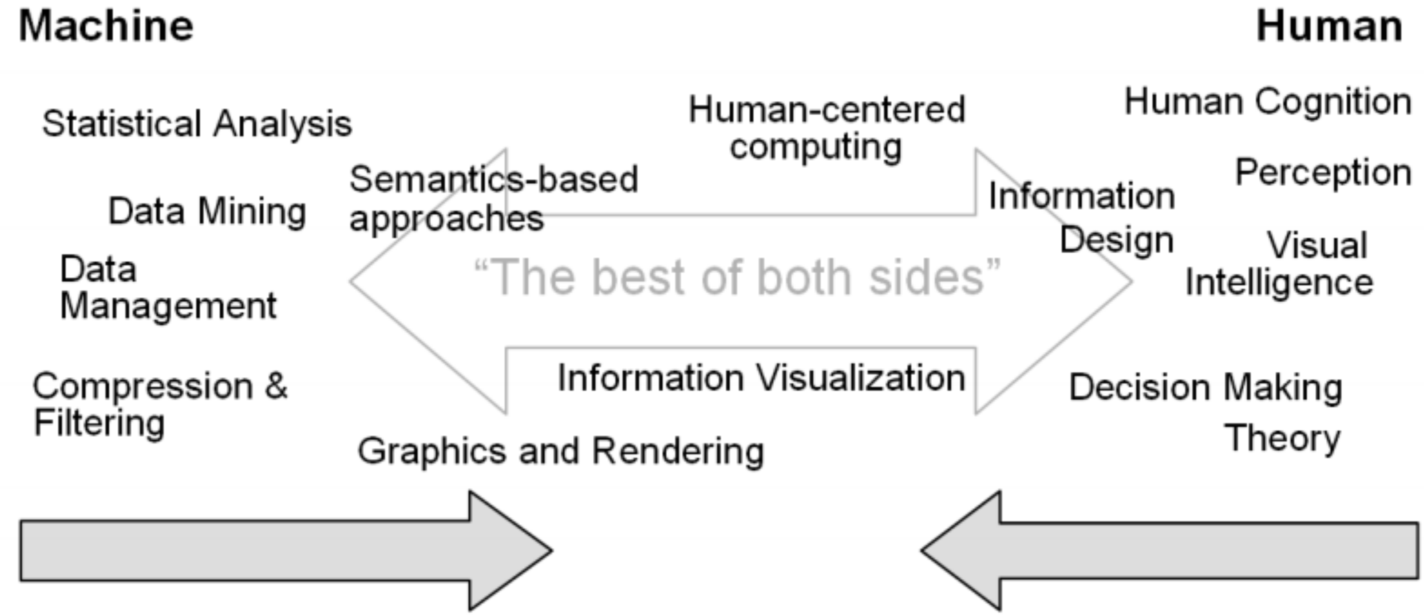
https://citytransit.uitp.org/

Information Visualization Reference Model





Visual Analytics

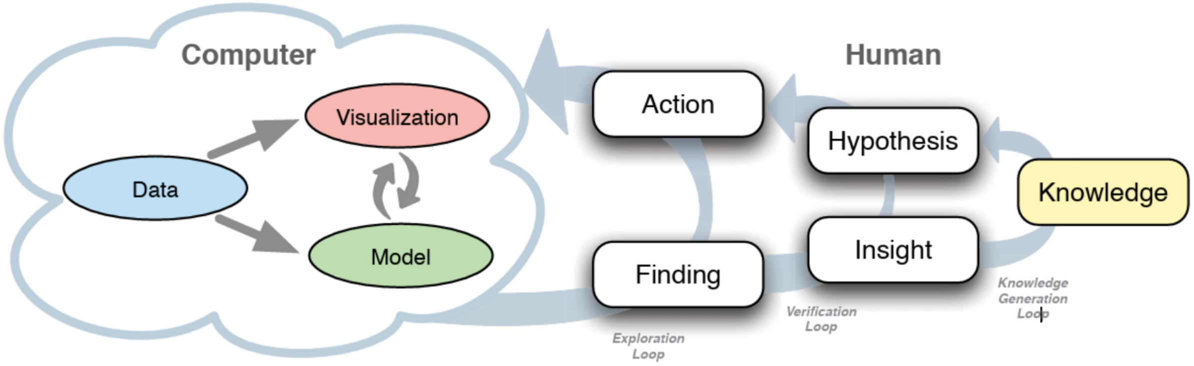


Keim, D., Andrienko, G., Fekete, J. D., Görg, C., Kohlhammer, J., & Melançon, G. (2008). Visual analytics: Definition, process, and challenges. In Information visualization (pp. 154-175). Springer, Berlin, Heidelberg.

1604 IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 20, NO. 12, DECEMBER 2014

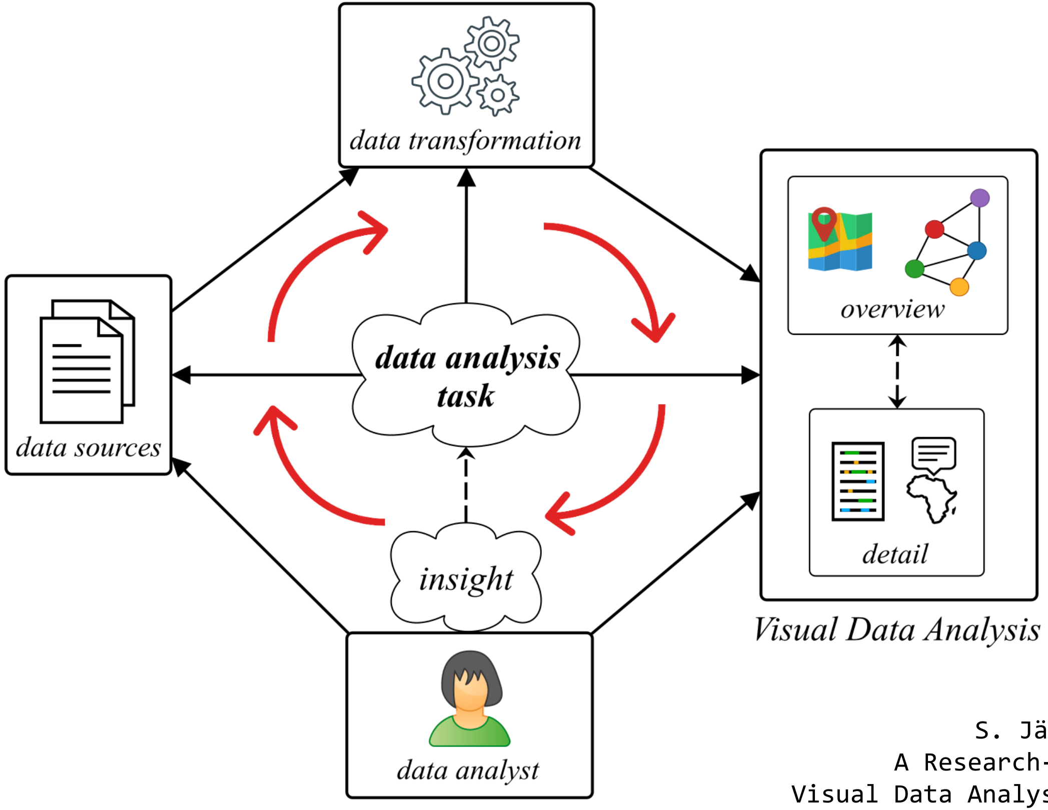
Knowledge Generation Model for Visual Analytics

Dominik Sacha, Andreas Stoffel, Florian Stoffel, Bum Chul Kwon, *Member, IEEE*,
Geoffrey Ellis and Daniel A. Keim, *Member, IEEE*



Sacha, D., Stoffel, A., Stoffel, F., Kwon, B. C., Ellis, G., & Keim, D. A. (2014). Knowledge generation model for visual analytics. IEEE transactions on visualization and computer graphics, 20(12), 1604-1613.

Visual Data Analysis



S. Jänicke (2020).
A Research-teaching Guide for
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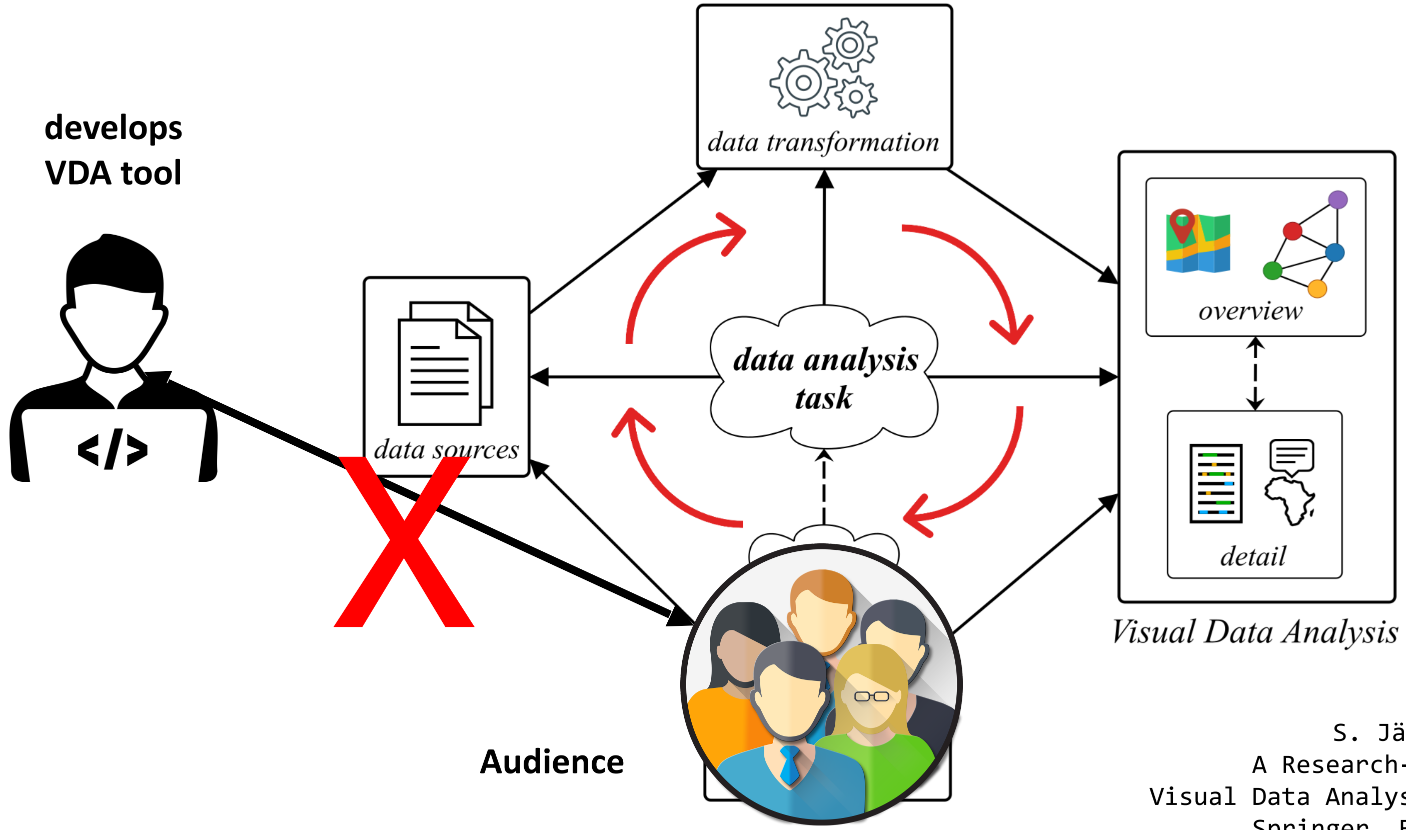
Springer, Berlin, Heidelberg.

Visual Data Analysis



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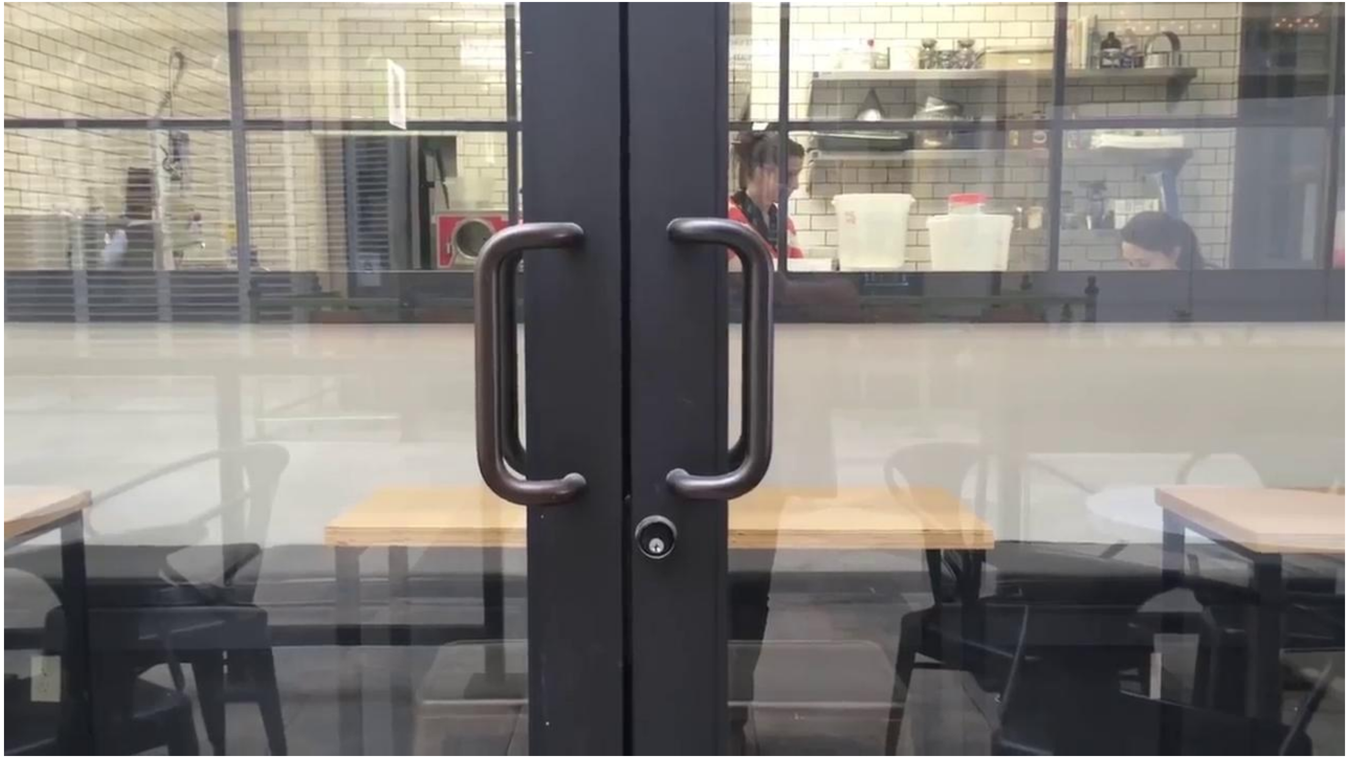
Mental Model: What Do We Know and What Do We Expect?

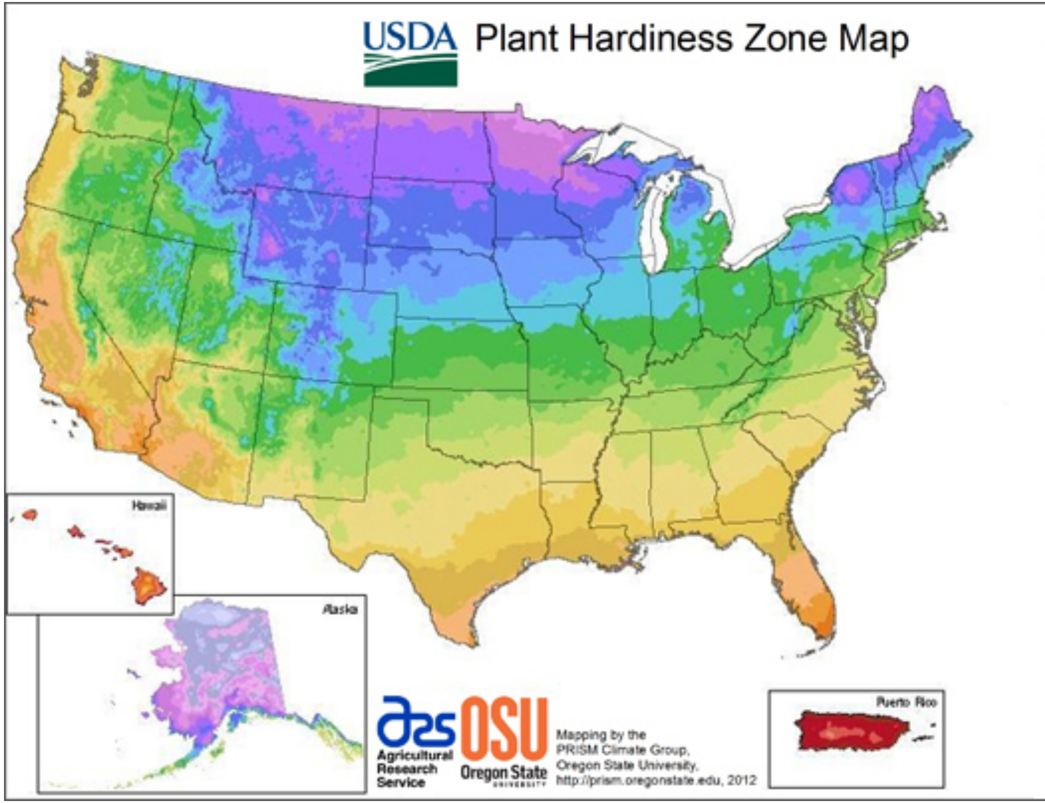
https://productcoalition.com/how-lean-usability-testing-can-improve-the-user-experience-7f3ef22392ea

Mental Model: What Do We Know and What Do We Expect?



https://www.youtube.com/watch?v=yY96hTb8WgI

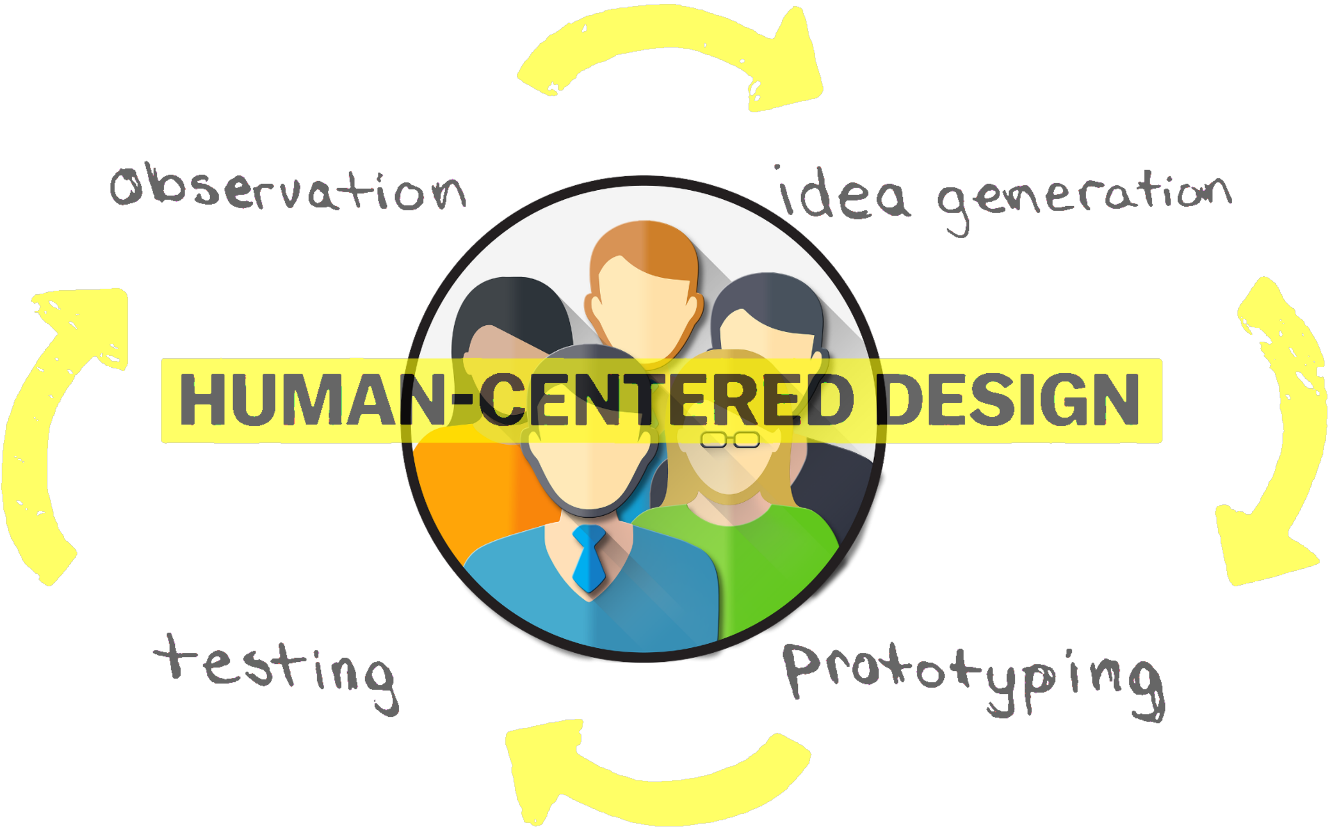
Mental Model: What Do We Know and What Do We Expect?



Temp (F)	Zone	Temp (C)
-60 to -55	1a	-51.1 to -48.3
-55 to -50	1b	-48.3 to -45.6
-50 to -45	2a	-45.6 to -42.8
-45 to -40	2b	-42.8 to -40
-40 to -35	3a	-40 to -37.2
-35 to -30	3b	-37.2 to -34.4
-30 to -25	4a	-34.4 to -31.7
-25 to -20	4b	-31.7 to -28.9
-20 to -15	5a	-28.9 to -26.1
-15 to -10	5b	-26.1 to -23.3
-10 to -5	6a	-23.3 to -20.6
-5 to 0	6b	-20.6 to -17.8
0 to 5	7a	-17.8 to -15
5 to 10	7b	-15 to -12.2
10 to 15	8a	-12.2 to -9.4
15 to 20	8b	-9.4 to -6.7
20 to 25	9a	-6.7 to -3.9
25 to 30	9b	-3.9 to -1.1
30 to 35	10a	-1.1 to 1.7
35 to 40	10b	1.7 to 4.4
40 to 45	11a	4.4 to 7.2
45 to 50	11b	7.2 to 10
50 to 55	12a	10 to 12.8
55 to 60	12b	12.8 to 15.6
60 to 65	13a	15.6 to 18.3
65 to 70	13b	18.3 to 21.1

http://www.lawngrasses.com/images/maps/usda/usda-2012.jpg

Human-centered Design



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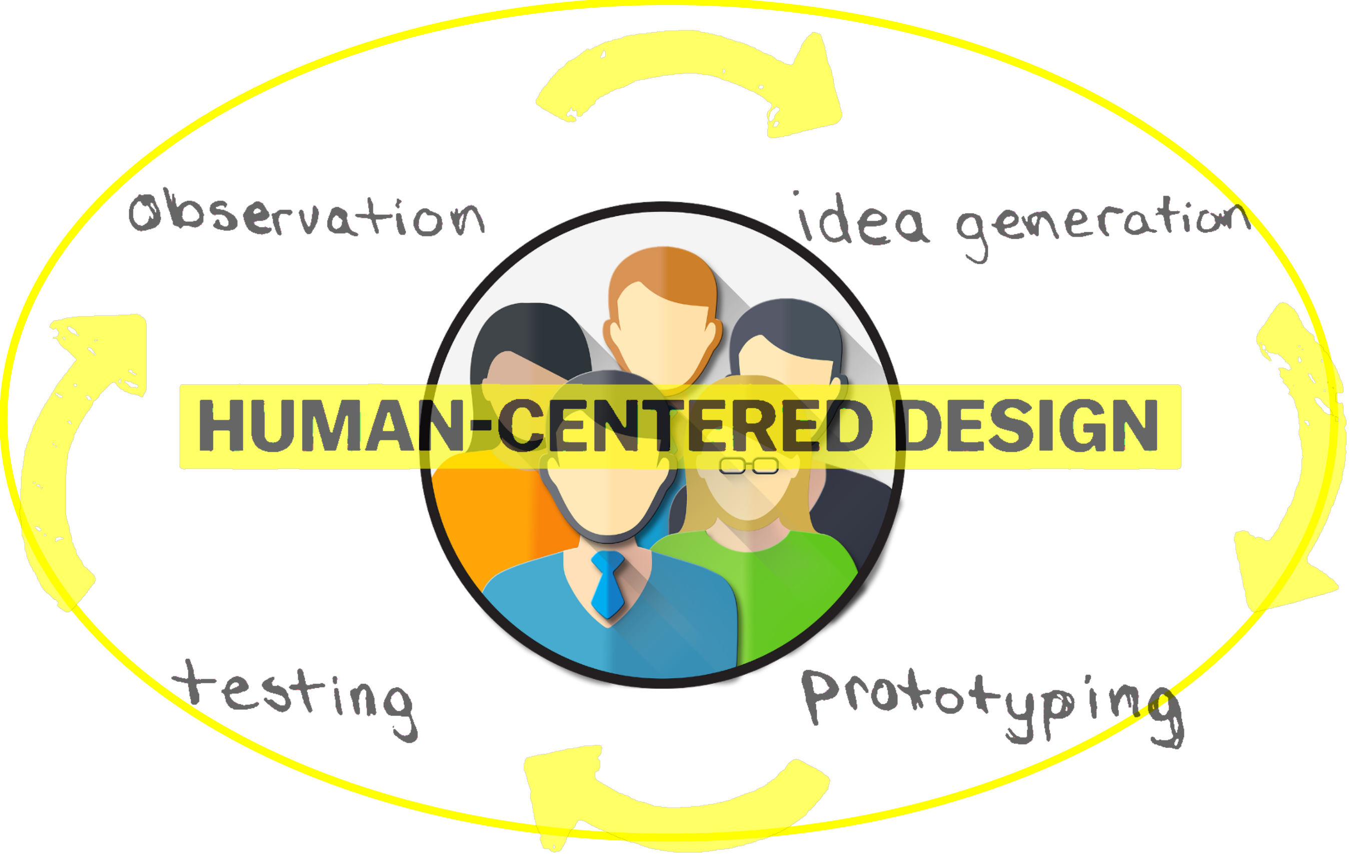
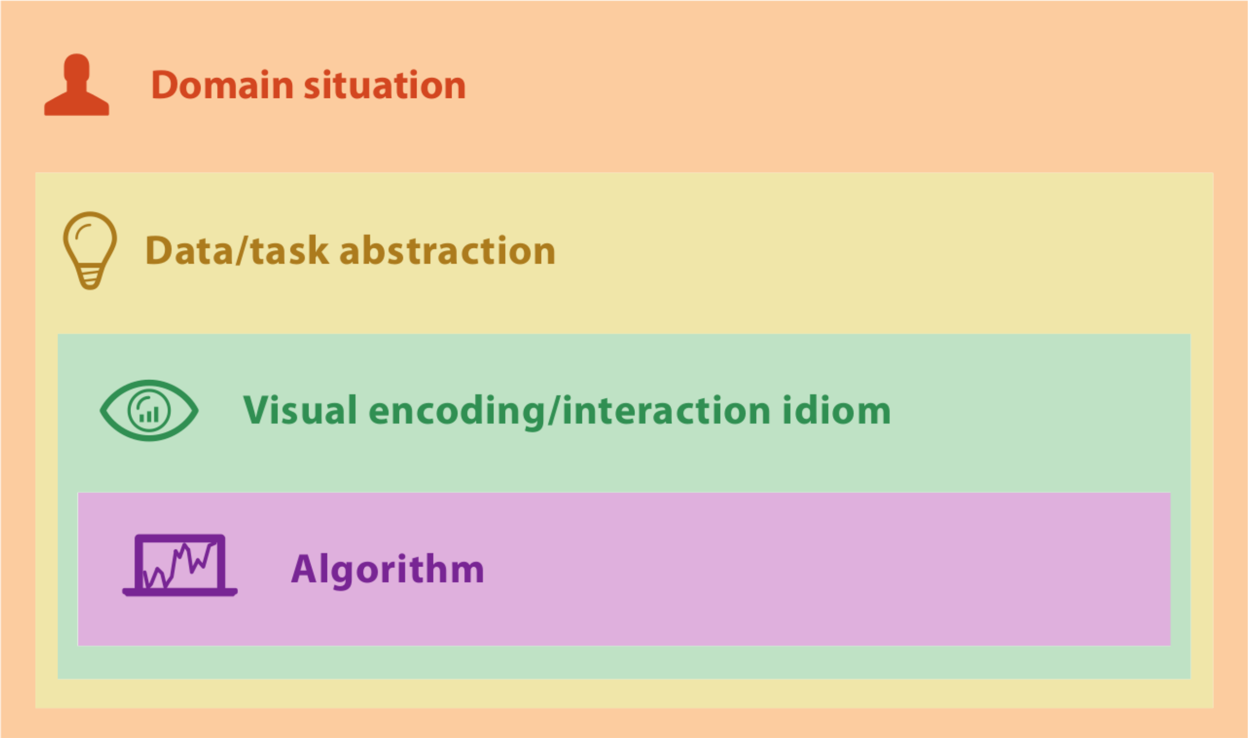


Nested Model for Visualization Design and Validation



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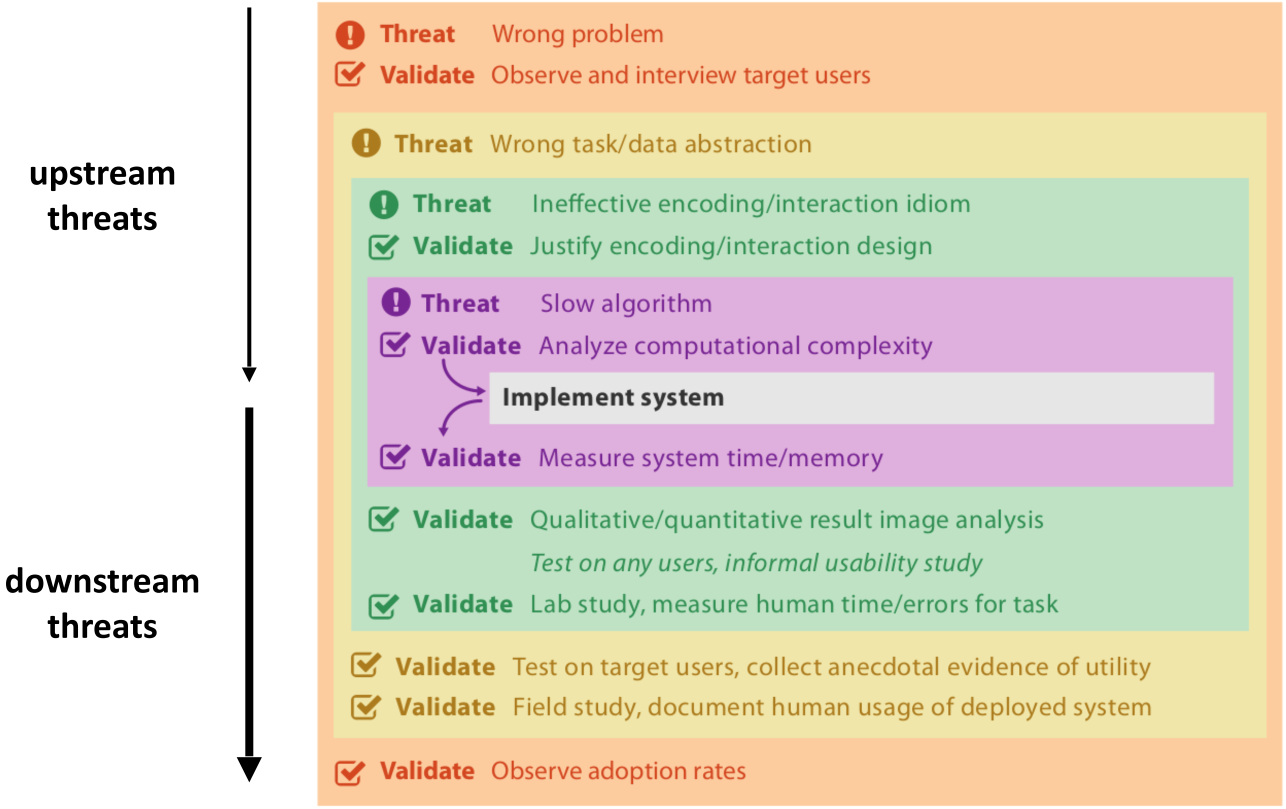

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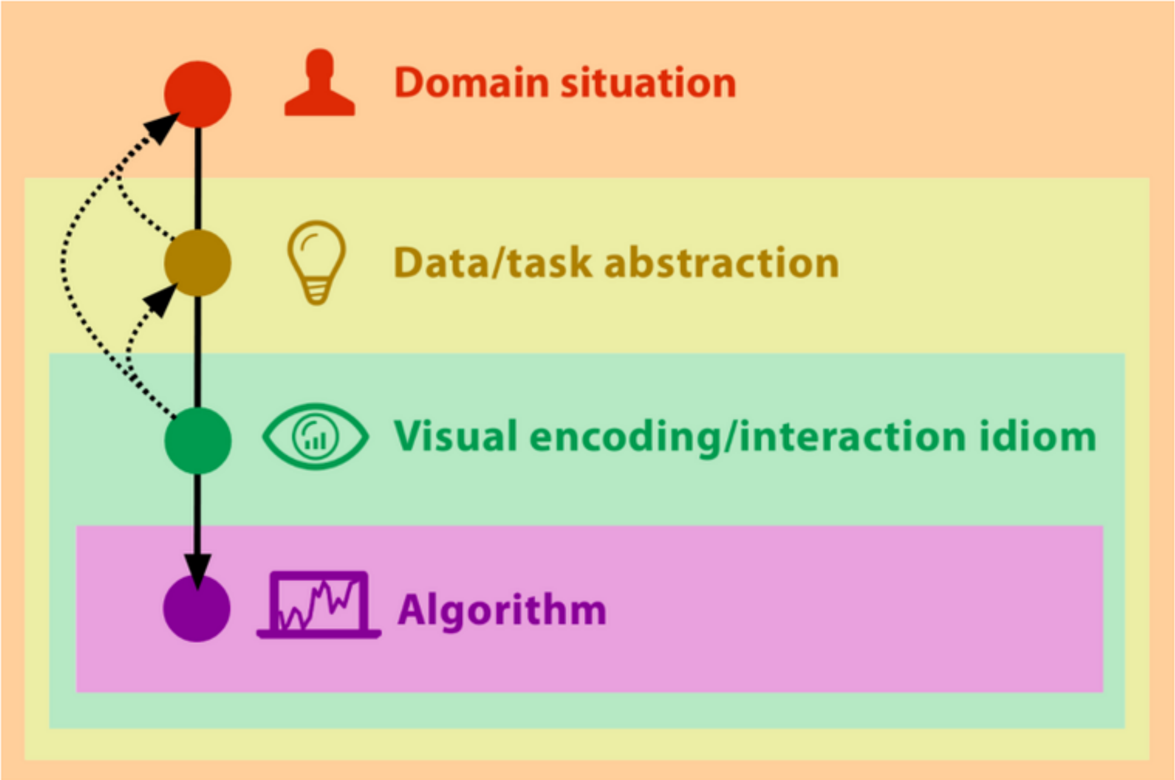
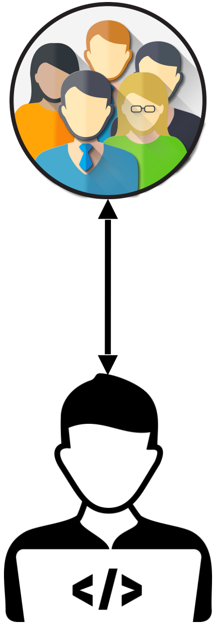
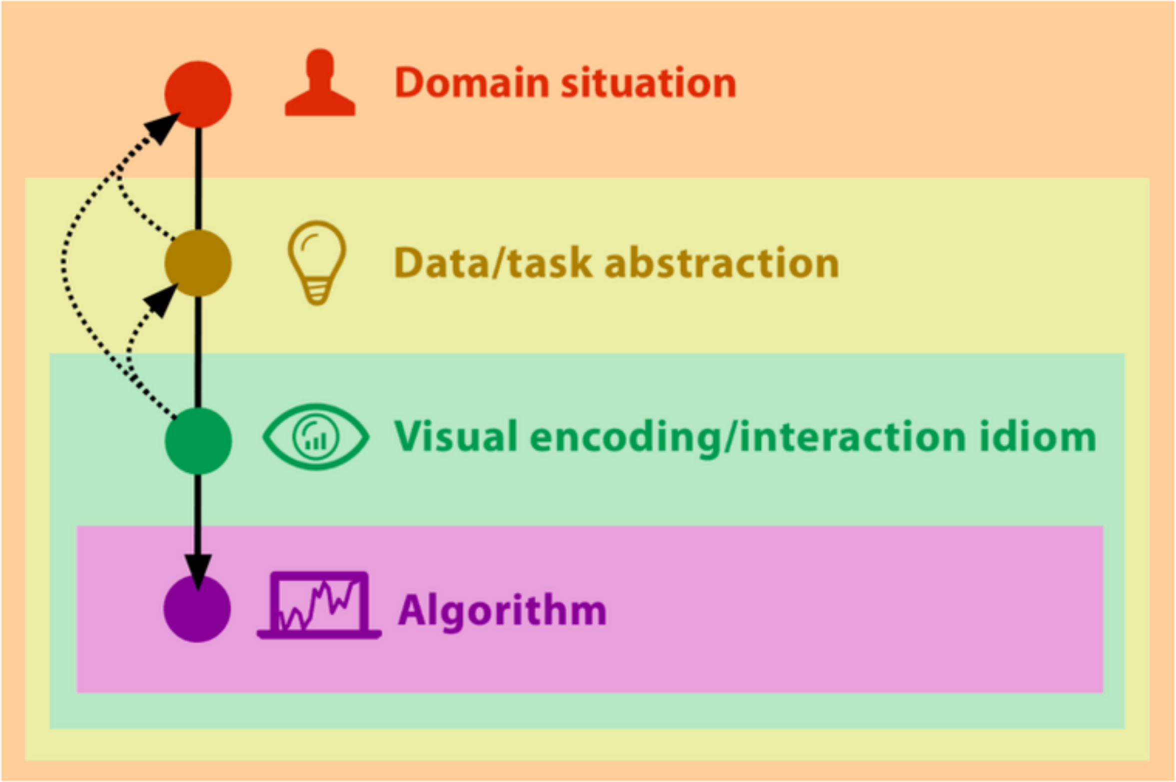
Participatory Visualization Design



Jänicke, S., Kaur, P., Kuzmicki, P., & Schmidt, J. (2020).
Participatory Visualization Design as an Approach to Minimize the Gap between Research and Application.

Participatory Visualization Design





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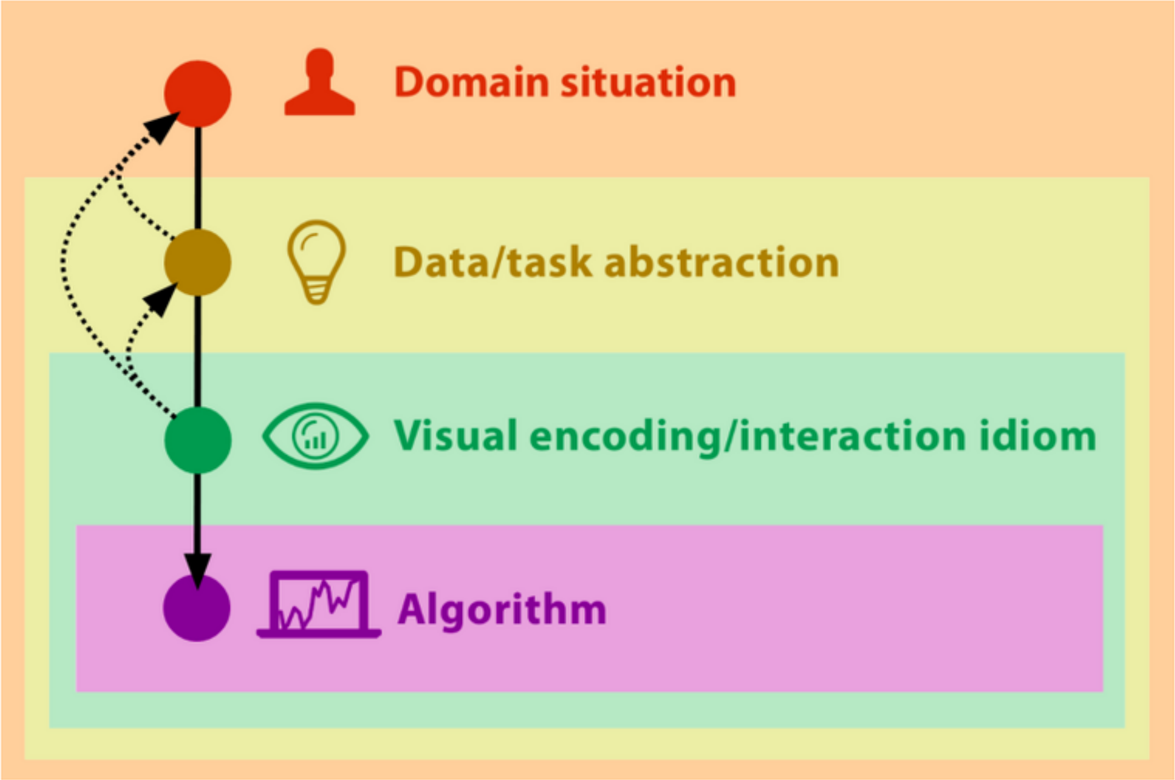
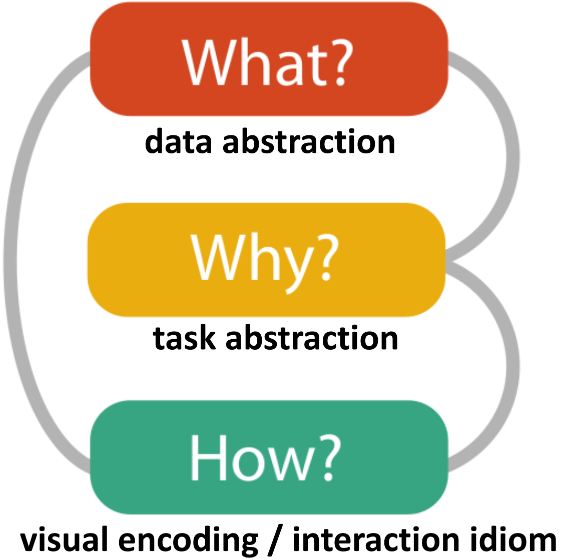




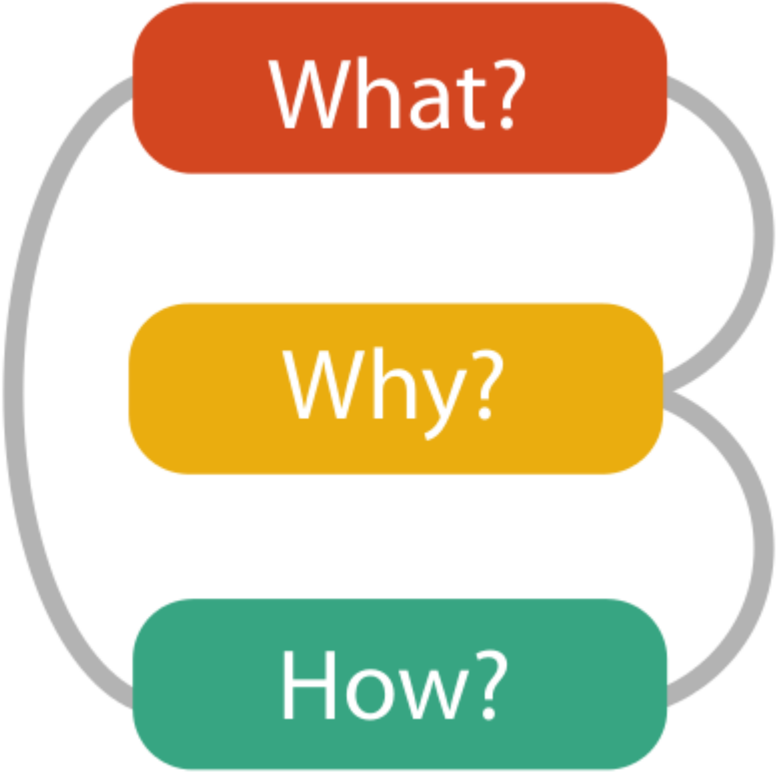
Participatory Visualization Design

Attributes
ordinal
numerical
categorial
relational

Availability
static
dynamic

Actions
analyze
present
search
derive
...

Visual Representations
tag clouds
maps, charts
timelines
...



Data types
entities
relations
attributes

Data set types
2D tables
nD tables
networks

Targets
trends
outlier
features
distributions
correlations
...

Interactions
select
filter
navigate
...

Takeaways

visualization design is made for  with a domain-specific  related to domain-specific 

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What?

data abstraction serves to turn domain-specific into standardized data formats processible for visualization purposes

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Why?

task abstraction serves to turn domain-specific into generic user tasks for which a visualization can be helpful

Takeaways

visualization design is made for  with a domain-specific  related to domain-specific 

What? data abstraction serves to turn domain-specific into standardized data formats processible for visualization purposes

Why? task abstraction serves to turn domain-specific into generic user tasks for which a visualization can be helpful

How? visual design includes a visual encoding strategy and an interaction idiom to generate an instrument for the user

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visualization design is made for  with a domain-specific  related to domain-specific 

What? data abstraction serves to turn domain-specific into standardized data formats processible for visualization purposes

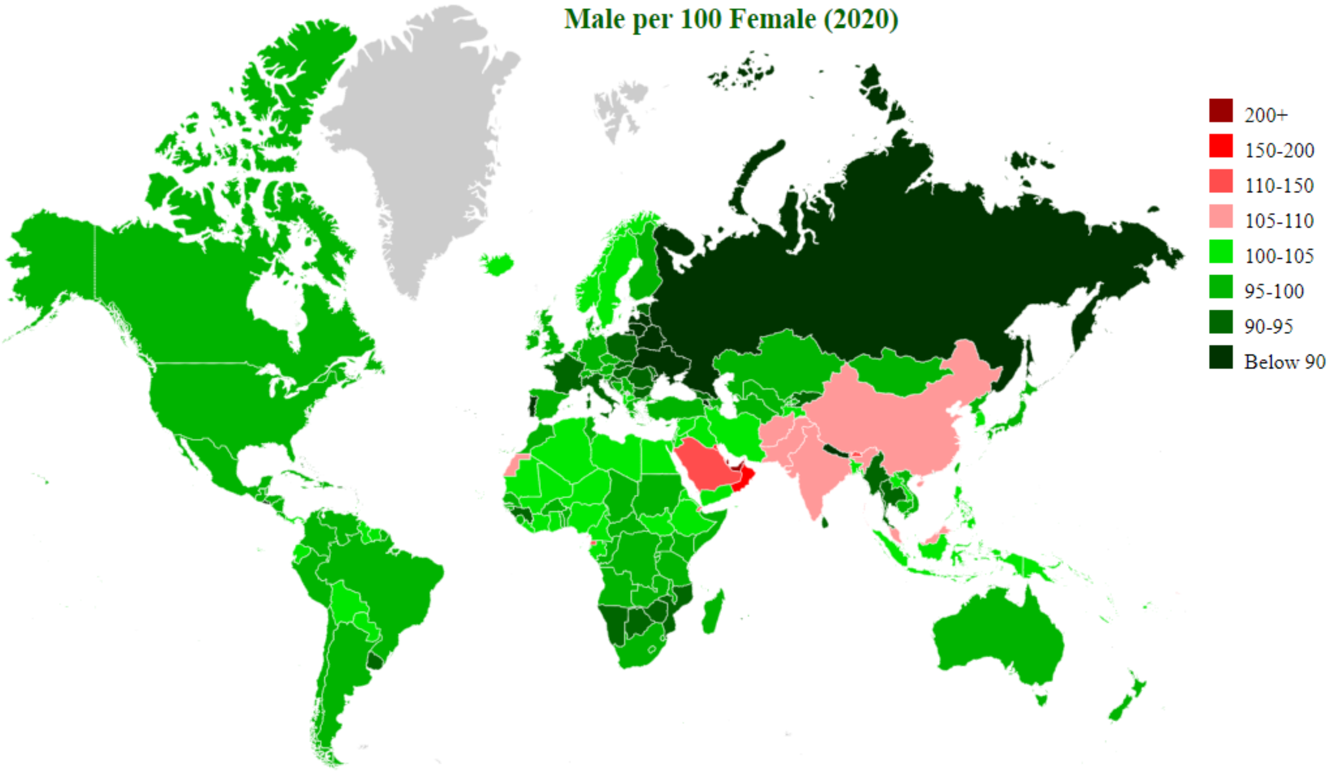
Why? task abstraction serves to turn domain-specific into generic user tasks for which a visualization can be helpful

How? visual design includes a visual encoding strategy and an interaction idiom to generate an instrument for the user

Participatory Visual Design: to develop a valuable tool that effectively helps to solve the intended task, many  concerning design considerations between 

and  are necessary

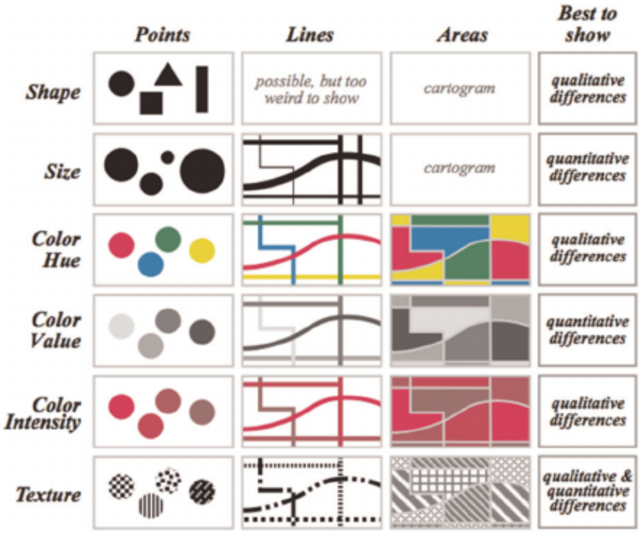
Task: What Seems to Be Contradictory to the Mental Model?



http://statisticstimes.com/demographics/countries-by-sex-ratio.php

How do we Map our Mental Model into a Visualization?

Mapping



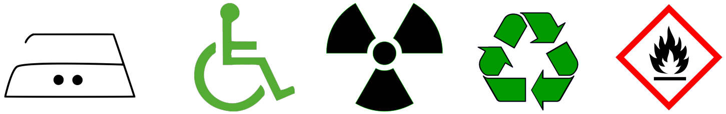
Semiology of Graphical Symbols

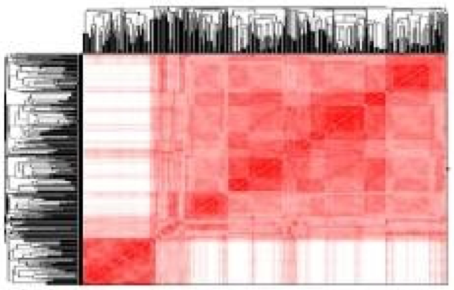
- Visualizations typically contain visual objects, graphical symbols
  - Semiology is the science of signs, graphical symbols, symbolism and communication
- Diagrams, networks, maps, plots all contain graphical symbols
- Good visualizations need well designed symbols and an understanding how they are perceived
  - Consider the common interpretation and semantic of symbols from everyday life

Symbol with obvious meaning

Symbols and Visualizations

- Some common symbols are universally recognizable
  - Perceived in one step through clearly associated meaning
  - May be preattentively recognizable
- Complex representations need multiple steps for understanding
  - Identify major graphical image elements
  - Identify relationships between them
- Human visual perception driven by physical interpretation
  - Visualization must have easily interpretable spatial (x,y,z) dimensions
- Any pattern, variation or order in the image must imply a pattern, variation or order in the d
  - Otherwise it is an artifact of the presentation



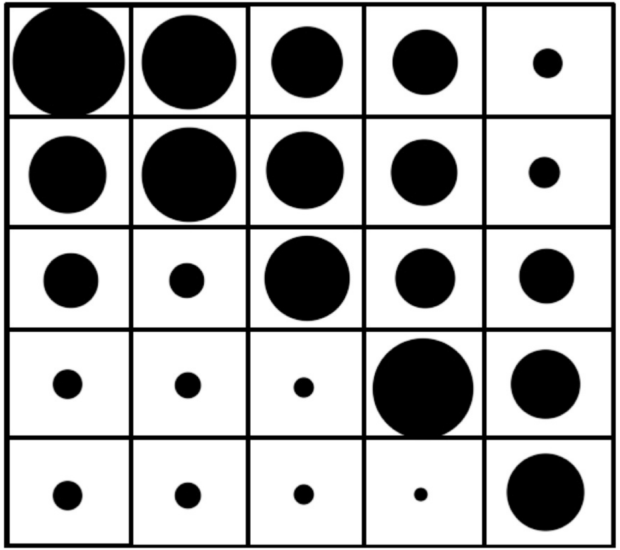


similarity in data ⇔ visual similarity of corresponding symbols

order between data items ⇔ visual order between corresponding symbols

Features of Graphics

- Graphics have three or more dimensions
  - x,y-position in (image) plane and size of symbol
- A graphic can contain large numbers of elements, themselves potentially forming a texture
  - Desired or unwanted effect
- Groups of objects may be recognized preattentively
  - Followed by cognitively characterizing these groups



Matrix representation of strengths of relationships between nodes in a graph

Marks

- A mark is made to represent some information other than itself
  - Represents a data point in a visualization
- Marks can be
  - **Points** are dimensionless locations on the plane, represented by signs that obviously need to have some size, shape or color for visualization
  - **Lines** represent information with a certain length, but no area and therefore no width, and are visualized by signs of some thickness
  - **Areas** have a length and a width and therefore a two-dimensional size
  - **Surfaces** are areas in a three-dimensional space, but with no thickness
  - **Volumes** have a length, a width and a depth, and are thus truly (filling) three-dimensional (space)

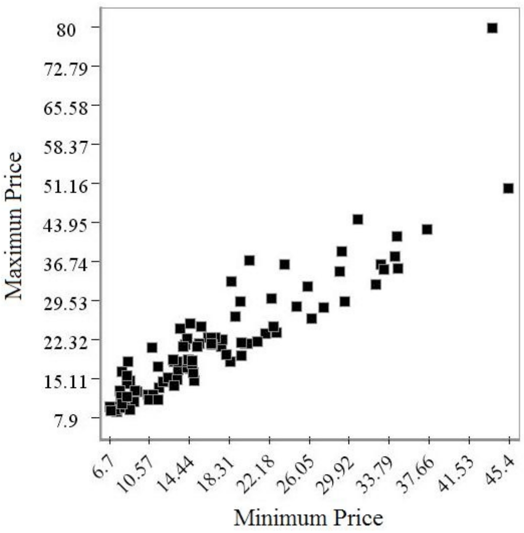
The Eight Visual Variables

1. Position
2. Shape (graphical primitive)
3. Size (length, area and volume)
4. Brightness
5. Color
6. Orientation
7. Texture
8. Motion

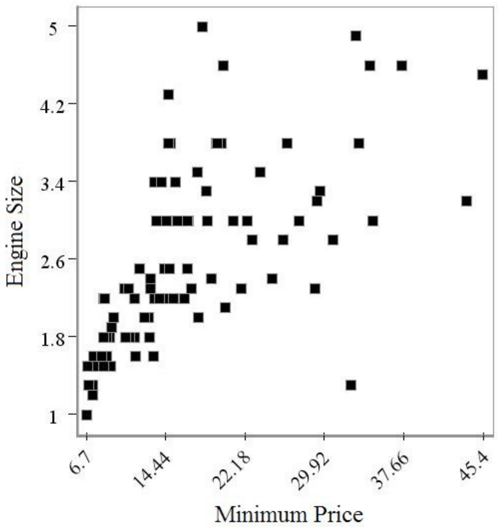
- Effects of Visual Variables

1. Position

- 1-, 2- or 3-dimensional
- Most important variable with greatest visual impact
- Space and pixels are limited
  - Cannot map multimillion data elements to unique positions on typical screens
- Linear and logarithmic scales can be applied
  - Consider distortion effects
- Mapping of two or more variables is generally a projection
  - From higher dimensional data to display space
    - eventually the image plane



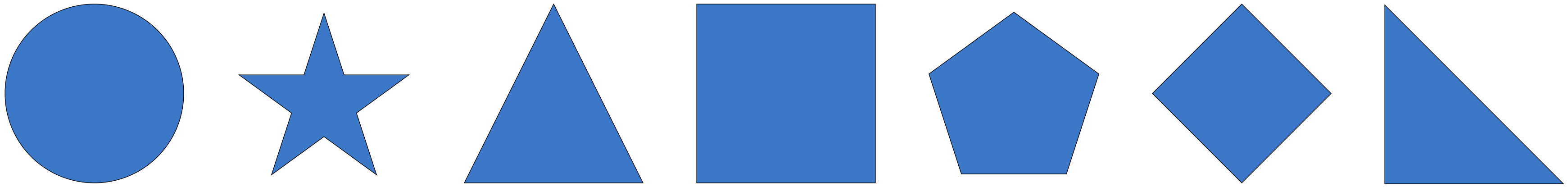
Display minimum price versus maximum price for cars with a 1993 model year. The spread of points appears to indicate a linear relationship between minimum and maximum price



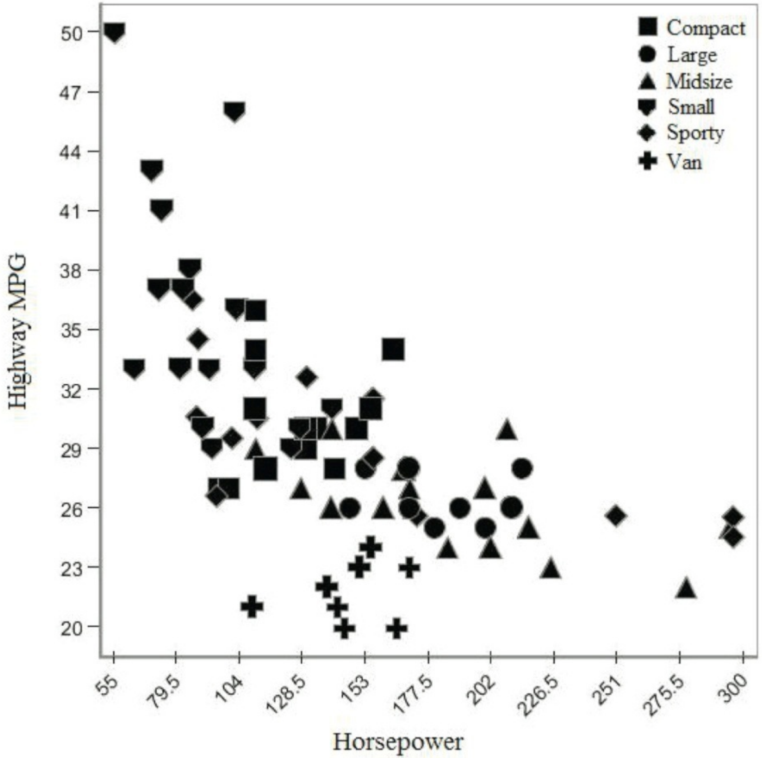
Compares minimum price with engine size for the 1993 cars data set. There does not appear to be a strong relationship between these two variables.

2. Shape

- The shape or *mark* can take on various forms
  - Points, lines, areas, volumes, complex 2D/3D shapes, symbols, letters, words
  - Care has to be taken with respect to differences in their relative appearance
    - covered area may be perceived as strength or some ordinal value
- Differentiability between marks important
  - Qualitative separation of data variable values
  - Hundreds or thousands of marks cannot be distinguished effortlessly

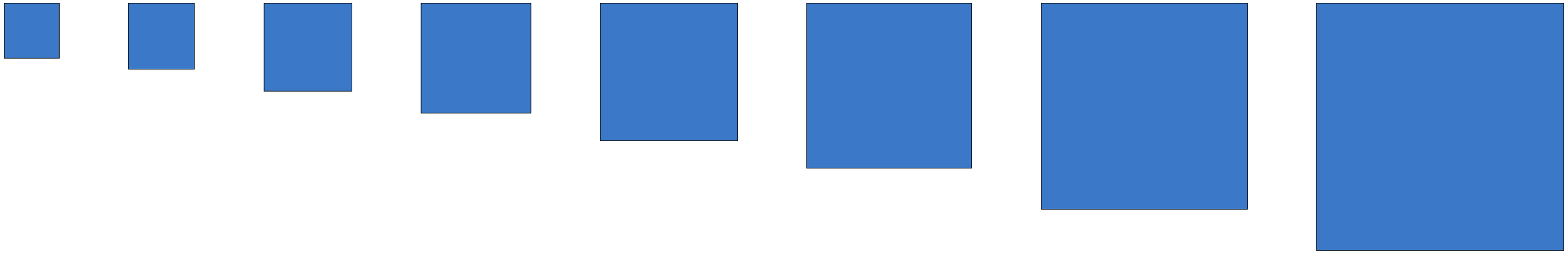


- Easy discernible marks
  - At least in low-density low-overlap areas
- Example: the mark is used to separate categories on top of a relational scatterplot
  - Comparing engine power and highway mileage for car types
  - Clusters are visible, as well as some outliers

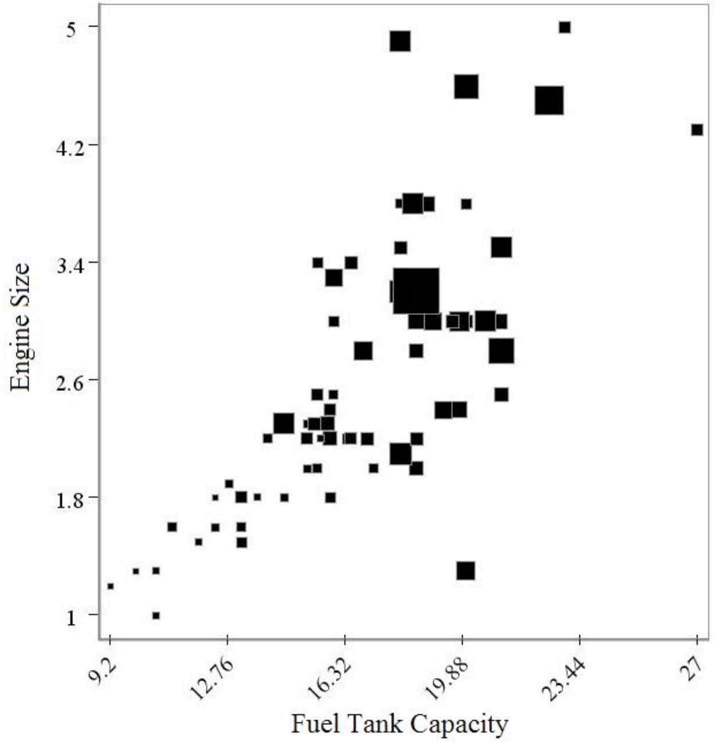


3. Size (Length, Area and Volume)

- Varying the size of the graphical element, mark is not mandatory
  - Position and mark are required though
- Additional attributes can be mapped to additional visual variables
- Size supports gradual relative ordering of data variables
- Quantitative separation aspect of size diminishes with sufficiently large area

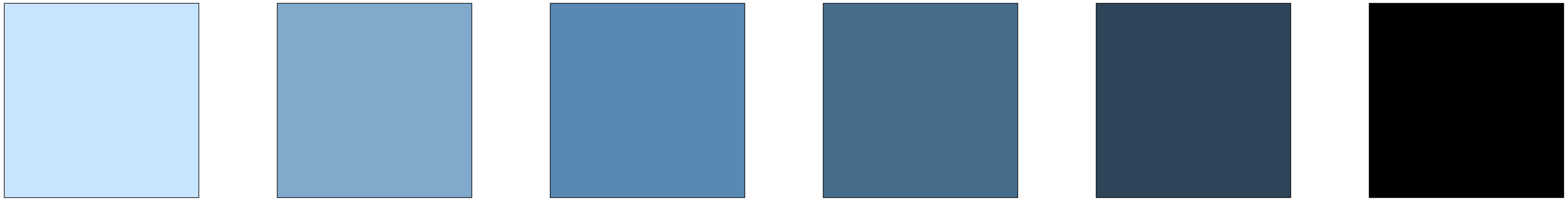


- Visualization of car models showing engine size versus fuel tank capacity. Size is mapped to maximum of price charged.
(larger is more expensive)



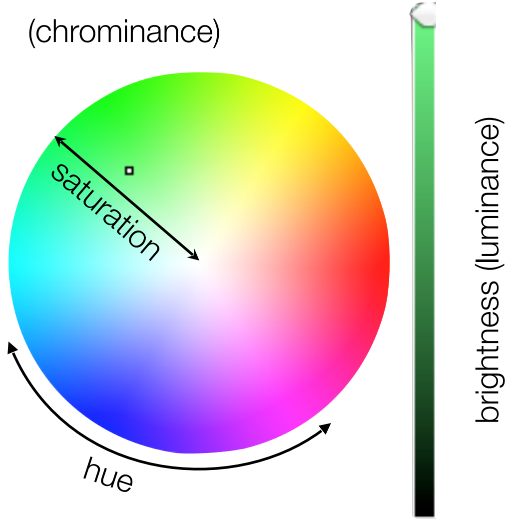
4. Brightness or Luminance

- Modify marks according to additional data variable
- Limited range of perceptually distinguishable brightness settings
  - Less critical for relative, in contrast to absolute differences of large interval and continuous data values
  - Reduced brightness scale can accurately identify data categories
- Perceptually linear brightness scales recommended
  - Use of gamma correction principle



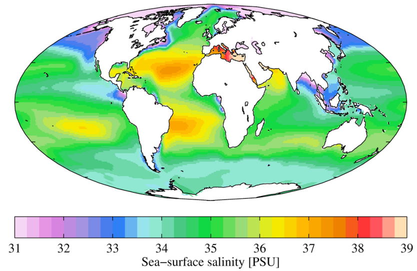
5. Color

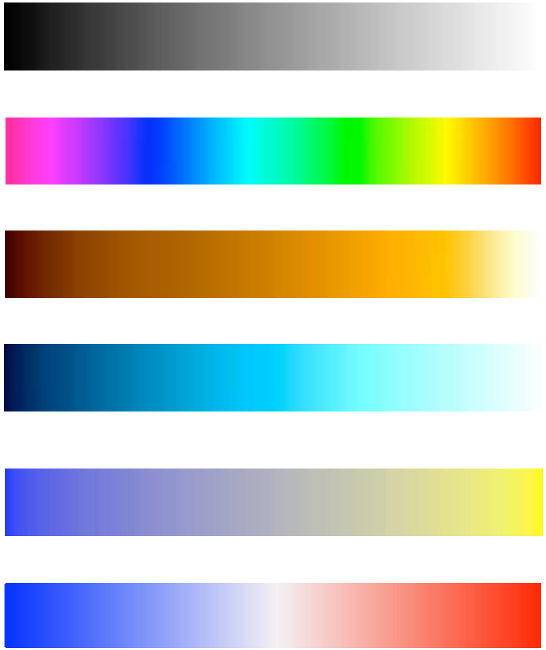
- Color can be specified using different color models
    - Factoring out brightness (luminance), color can be specified using two (chrominance) parameters
- Hue defines the color's dominant wavelength in the visible spectrum
- Saturation indicates the intensity of the color
    - Relative to grey (or white for maximal brightness)
- Colormaps can be used to encode continuous/ discrete (ordinal) and categorical (nominal) data variables



Common Colormaps

- Use color gradients which are intuitive and distinguishable
- Use discretized color palettes that are perceptually well separated





standard linear gray scale

rainbow

heated

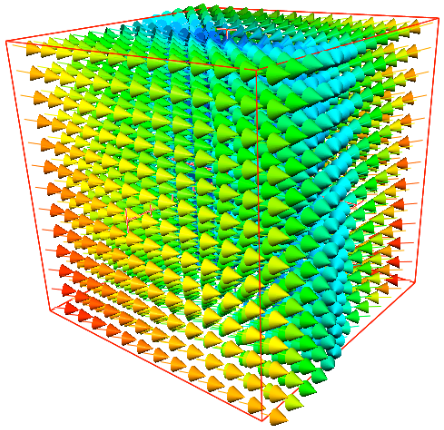
blue to cyan

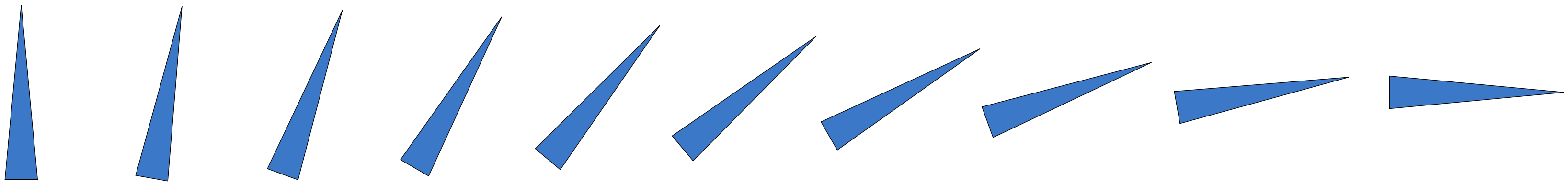
blue to yellow

cold-hot/negative-positive

6. Orientation

- Principle component of elongated marks with a clear symmetry axis
  - Rotational symmetry of mark shapes, e.g. a for a star, destroys orientability
- Rotational orientation supports relative ordering
  - Especially within a constrained but continuous range or interval
  - Limited absolute differentiability
    - cannot distinguish between 360 rotations



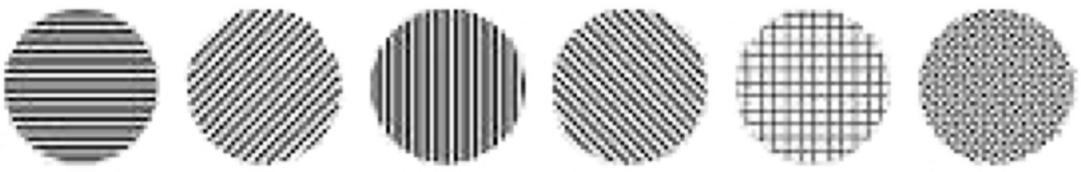


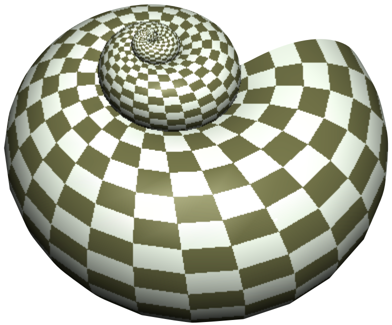
7. Texture

- Similar to the mark as visual variable of a point sample, but also applicable over large spatial extents
  - Can differentiate qualitatively between categories
  - But may also provide relative quantitative information
    - e.g. density or spacing of patterns, color range

Six possible example textures that could be used to identify different data value categories.

- Can also improve perception of 3D surfaces and shapes

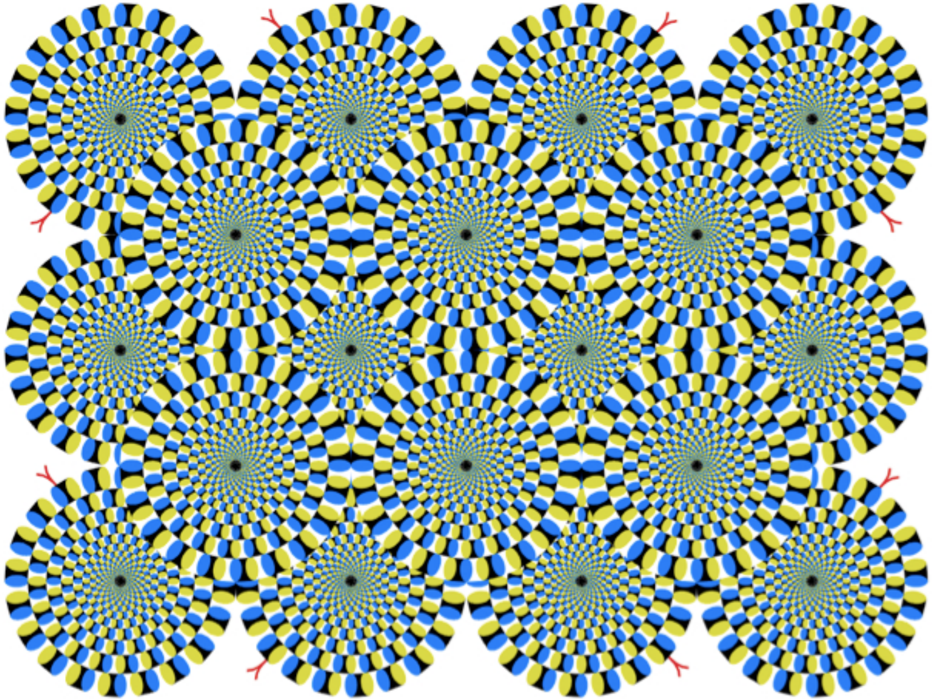




8. Motion

- Motion as change over time can be applied to all visual variables
  - Dynamically animated position
  - Deformation or transformation of mark shape
  - Change of color, size, orientation
- Rate of change can be indicative of a data variable
  - E.g. jittering of position or flashing (change of opacity)

Optical Illusion

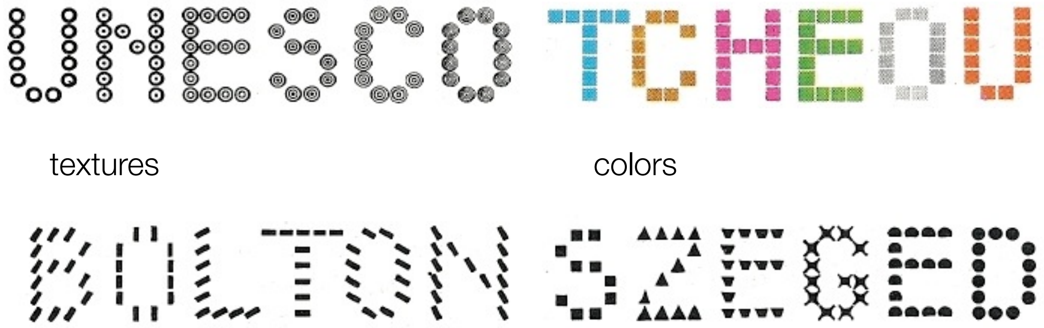


Effects of Visual Variables

- Selective visual variables
  - Can divide different data values into distinguished groups (for visualizing nominal values)
    - Color
    - Size (length, area, volume)
    - Brightness
    - Sample/texture
    - Direction/orientation
    - Shape
- Ordinal visual variables
  - Support easy ordering of data values (for visualizing ordinal and quantitative data)
    - Size (length, area, volume)
    - Brightness
    - Sample/texture
    - Direction/orientation

Effects of Visual Variables

- Associative visual variables
  - Associative if it does not affect the visibility of other dimensions (for visualizing nominal values)
    - Sample/texture
    - Color
    - Direction/orientation
    - Shape



textures colors

direction shape

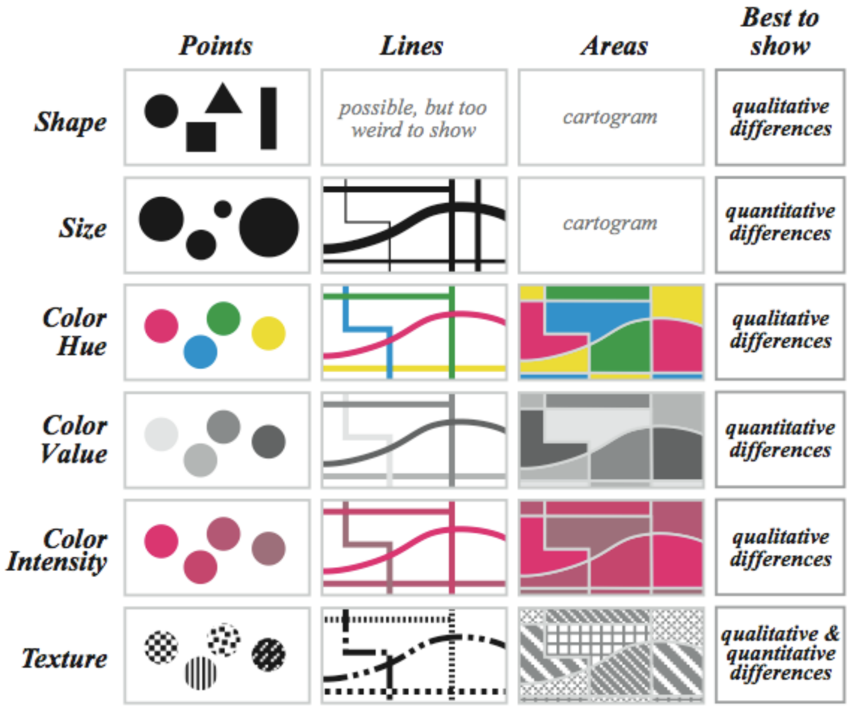
Effects of Visual Variables

- Proportional visual variables
  - Allow direct association and comparison of relative size (for visualizing ordinal and quantitative data)
    - Size (length, area, volume)
    - Direction/orientation
    - Brightness
- Separating visual variables
  - All data elements are visible and separable
    - Sample/texture
    - Color
    - Direction/orientation
    - Shape

Separating texture

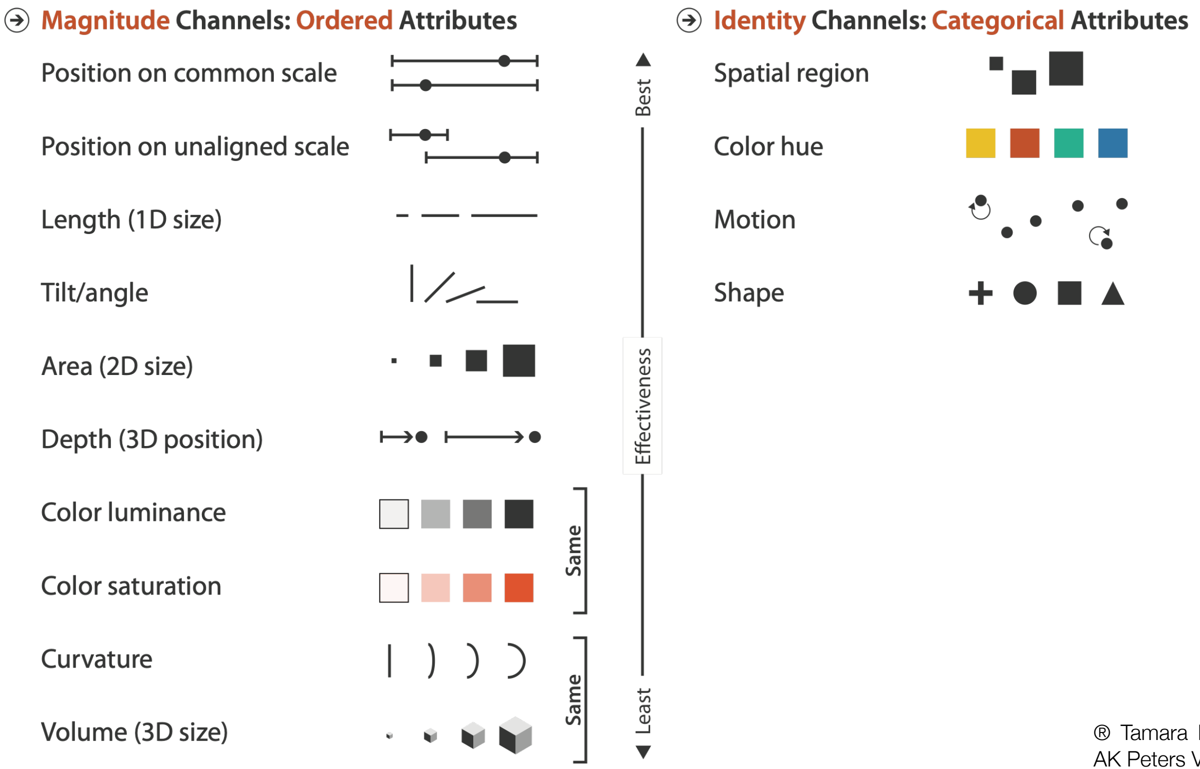
- Effectiveness for showing qualitative or quantitative differences according to Bertin
  - Summarized in *[Making Maps: A Visual Guide to Map Design for GIS](#)* by John Krygier and Denis Wood
  - Krygier and Wood also show how to represent the attribute through points, lines, or areas

- Study of visual variables has been traditionally important in cartography



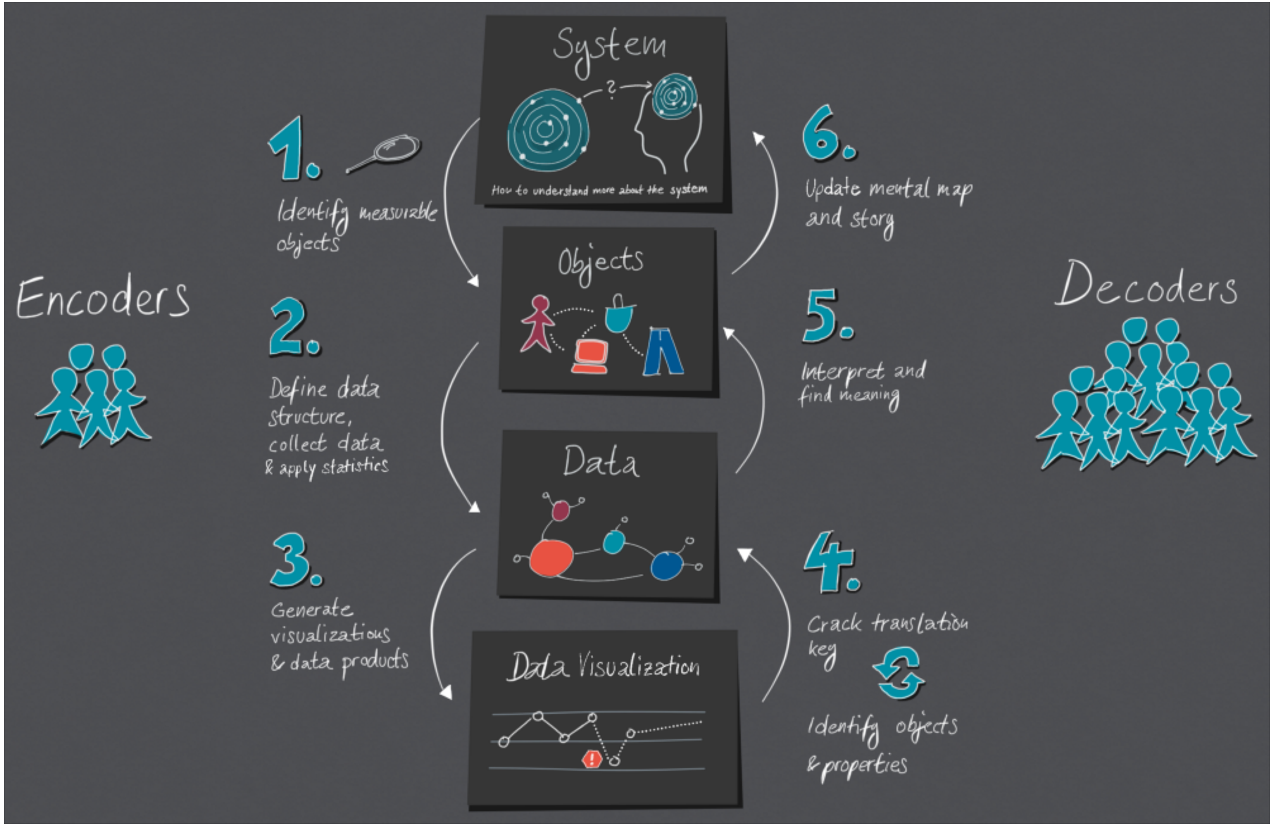
http://understandinggraphics.com/visualizations/information-display-tips/

Marks and Channels



® Tamara Munzner. Visualization Analysis and Design. AK Peters Visualization Series, CRC Press, 2014.

The Cycle of Encoding and Decoding

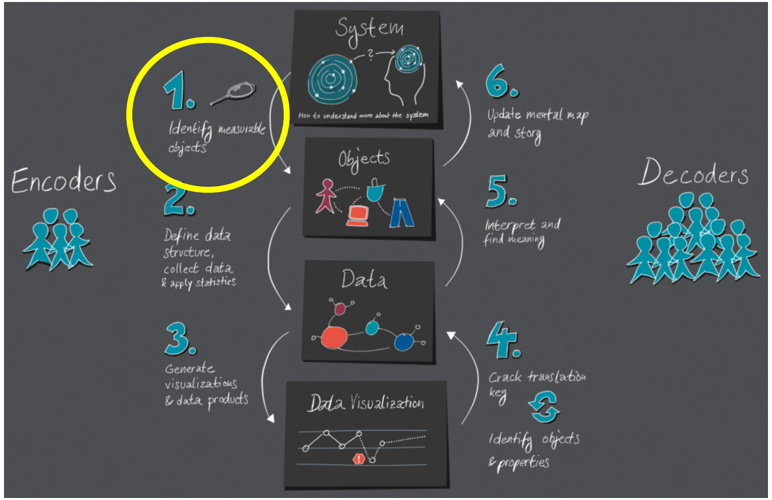


https://medium.com/nightingale/the-cycle-of-encoding-and-decoding-f3ff17010631

The Cycle of Encoding and Decoding

ENCODING

done by designer



1. Identify measurable objects

“data” is part of a complex system
we require a “mental map” of this system
What (objects, properties, events, relations) could be interesting for the user?
mental map should be complete and accurate

The Cycle of Encoding and Decoding

ENCODING

done by designer

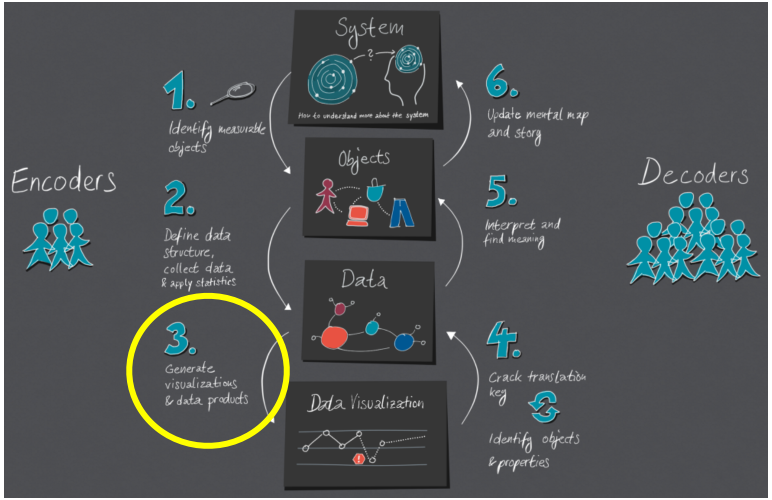
2. Define the data structure, collect data and apply statistics

create data model
not all information can be captured as data
collect data
transform raw data into data model (1st translation)

The Cycle of Encoding and Decoding

ENCODING

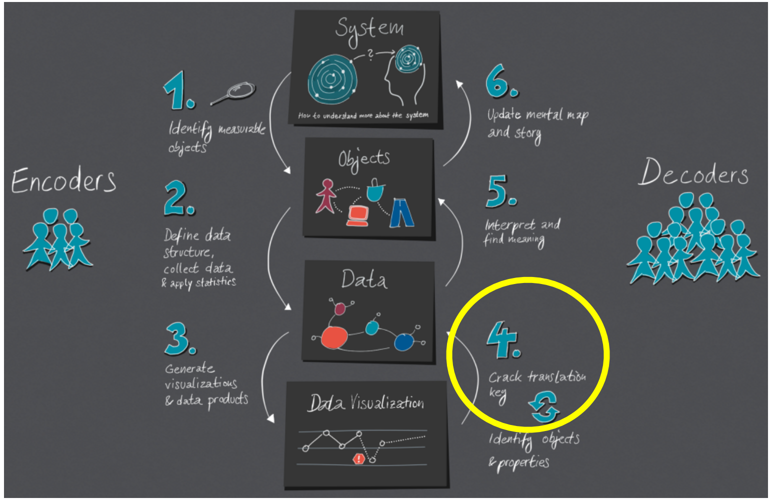
done by designer



3. Generate visualizations and data products

create visualization
transform data into visual representation (2nd translation)
focus on message to convey and the needs of the users
consider perceptive skills

The Cycle of Encoding and Decoding



DECODING

done by user

4. Crack translation key and identify objects and properties

understand meaning of visuals with reference to underlying data
What do lines, points, positions, and colors mean?
What do labels and legends say?
understand underlying data structure

DECODING

done by user

5. Interpret and find meaning

understand real world objects represented by the data
What properties/relations do they have? What patterns exist?
decoder wants to learn about the inner workings of the underlying system: *Why …?*
involves domain knowledge, i.e., the mental map of the system

The Cycle of Encoding and Decoding

DECODING

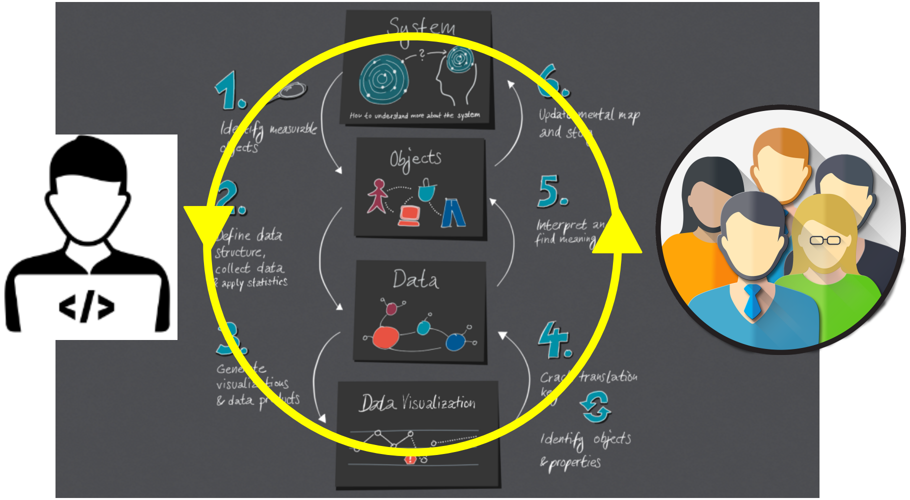
done by user

6. Update mental map and story

expand knowledge, improve mental map, ask further questions

"It is of utter importance that we as data encoders and creators of data visualizations make every effort to find out the exact job to be done of the product. If we design the data visualization product to fulfill the user's need, the knowledge extracted in this process actually becomes useful."

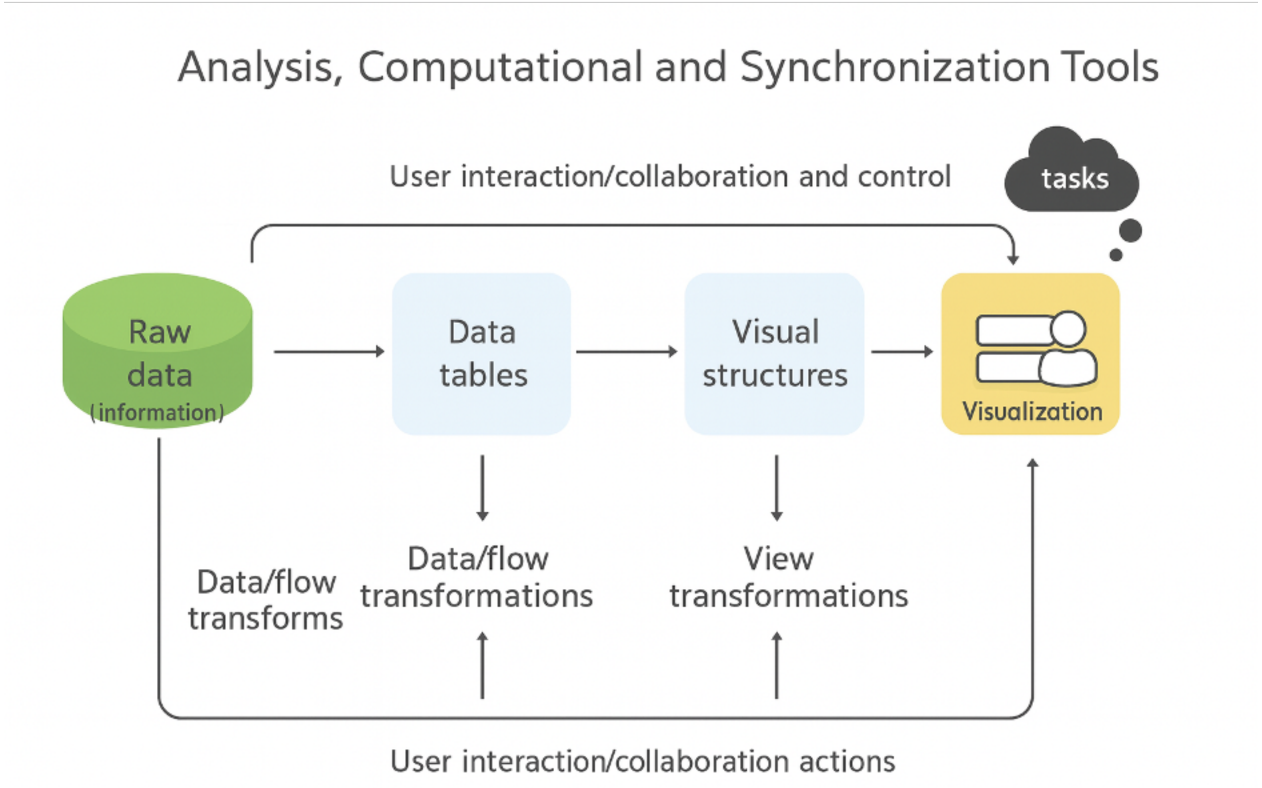
The Cycle of Encoding and Decoding



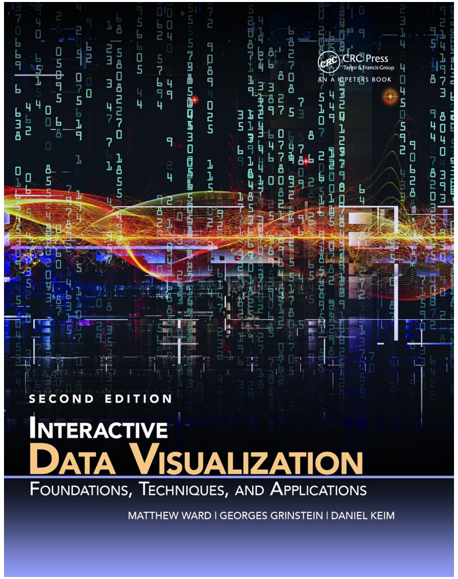
Decoder:
tries to connect visually exposed information to
what is already known (the current mental model of the system)

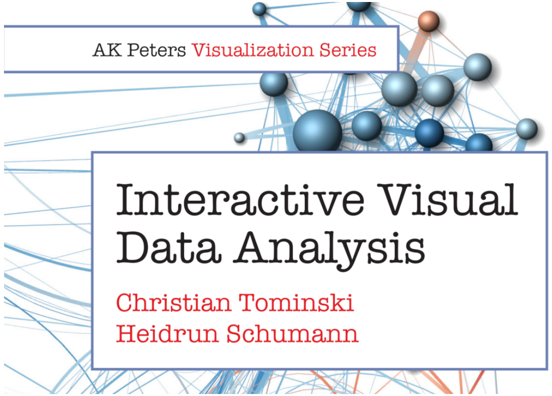
Encoder:
needs to ensure that audience has all the important bits of knowledge
to decode the data visualization

Visualization Pipeline



Some resources





Further Reading

Card, M. (1999). *Readings in information visualization: using vision to think*. Morgan Kaufmann.

Keim, D., Andrienko, G., Fekete, J. D., Görg, C., Kohlhammer, J., & Melançon, G. (2008). Visual analytics: Definition, process, and challenges. In Information visualization (pp. 154-175). Springer, Berlin, Heidelberg.

Münster, E. (2020). The Cycle of Encoding and Decoding: How does data visualization work?
https://medium.com/nightingale/the-cycle-of-encoding-and-decoding-f3ff17010631

Norman, D. (2013). The design of everyday things: Revised and expanded edition. Basic books.

Munzner, T. (2014). *Visualization analysis and design*. CRC press.
Chapter 4: Analysis: Four Levels for Validation

QUESTIONS?